What does a Computation actually Prove?

From Algorithmic Choice to Formal Ethical Certification

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Abstract

What can an observed algorithmic choice establish, and what conditions must the evidence satisfy for us to assert that one alternative is superior to another? This research distinguishes between the observed selection of an alternative, the relations that can be mathematically determined from partial evidence, and the guarantees required to justify a strict preference under a declared standard. This distinction allows us to specify what knowledge an algorithmic decision provides and which conclusions require additional evidence.

We present three logical-mathematical results in addition to the central characterization:

- First, an algorithmic selection does not, by itself, entail a demonstrated strict preference.
- Second, a single valid counterexample—a legal alternative whose attested value exceeds that of the selected one—suffices to refute the universal correctness of a configuration with respect to the declared domain and objective function.
- Third, consistent extension of the evidence preserves previously determined relations, provided that the same semantic conditions are maintained.

The main result concerns finite MIN/MAX evaluation expressions, including those that share leaf variables, provided that the unknown values can vary independently within closed, bounded real intervals: in that case, strict dominance of b over a in all completions compatible with the evidence holds if and only if the exact lower extremum of b exceeds the exact upper extremum of a.

The sufficiency of this separation is classical; we prove the converse by constructing a single compatible completion, by threshold clipping, in which b does not exceed a, without requiring disjoint subtrees. The proof relies on known algebraic properties of lattice polynomial functions, and we make no claim of historical priority over equivalent mathematical characterizations.

The contribution of this research lies in formulating and developing a problem of justifying algorithmic decisions from partial evidence, establishing its logical-mathematical conditions, and proving, within the declared model, an exact characterization of strict dominance. On the basis of these results, we propose a formal grounding for the epistemic and evidential dimension of Responsible AI (RAI), based on the distinction between the observed selection, the mathematically demonstrated conclusion, and the additional guarantees required for its computational certification.

A worked example with a twelve-leaf graph illustrates the result: five observed leaves suffice to determine which of the two alternatives is superior and, if any one of them is removed, the comparison is left open, as shown by two compatible completions with opposite outcomes, constructed explicitly. The result is limited to the independent-interval uncertainty model: it establishes no properties of any search algorithm and expressly separates what the evidence determines from what a computational procedure certifies.

A final section shows, through a countermodel conditioned on explicit normative premises, that computational correctness according to a criterion Φ does not by itself entail the moral justification of an action under a normative framework N independent of Φ. We do not assume any universally valid moral framework.

Keywords: algorithmic choice; partial evidence; MIN/MAX evaluation graphs; semantic entailment; countermodels; bound separation; demonstrated preference; Responsible AI (RAI); formal verification.

Introduction

In the preceding article, *What Does an Algorithmic Choice Demonstrate? A Formal Analysis of Evidence, Comparison, and Preference Attribution* (Valenzuela Alonso 2026b), we posed a deceptively simple question: "Does selecting A allow us to conclude that an algorithm prefers A to B?" Its thesis can be summarized as follows: choice is an observable fact; demonstrated preference is a conclusion that requires additional conditions on the evidence.

This result establishes a precise logical limit on the inferences that can be drawn from an observed algorithmic decision. In this research we formulate and prove Proposition 2, which establishes this non-entailment within the class $\mathcal{K}_{sel}$ of structures satisfying the definitions of comparison resolution and demonstrated preference, without imposing additional axioms that would necessarily link selection to an admissible witness:

$$Selected_h(b) \;\not\models_{K_{sel}}\; DP_h^C(b,a)$$

Observing that alternative b was selected in history h does not allow us to conclude, from that fact alone, that there exists a demonstrated strict preference for b over a under the declared standard C. The non-entailment is relative to that class: it would break down if a premise linking selection to an admissible witness were added.

1 The question

This second study examines what happens when the evidence is partial:

Under what conditions does the available evidence allow us to prove that *b* is strictly superior to *a*, and what are the limits of what that proof permits us to conclude?

We distinguish three notions that are often conflated when analyzing algorithmic decisions:

- the fact that an alternative has been selected;
- the semantic determination of a comparison from partial evidence;
- the attribution of a demonstrated preference under a declared standard.

Based on these distinctions, we present two auxiliary results:

- A single valid counterexample refutes universal correctness.

If, in some instance x_0 of the domain D, there exists a legal action a whose value exceeds that of the chosen action $A(x_0)$, the configured implementation A is not universally correct ($UC(A, D, V)$ is false in D):

$$x_0 \in D \;\wedge\; a \in L(x_0) \;\wedge\; V(x_0, a) > V(x_0, A(x_0)) \;\Rightarrow\; \neg UC(A, D, V).$$

- Consistent extension of the evidence preserves determination.

Let $E \subseteq E'$ be two evidence sets interpreted under the same semantics (the same class of models and the same contracts), with E' consistent, that is, with a nonempty set of compatible completions. Then every relation determined by E is determined by E':

$$E \subseteq E' \Rightarrow \mathrm{Compat}(E') \subseteq \mathrm{Compat}(E), \qquad E \models R \Rightarrow E' \models R.$$

The inclusion $\mathrm{Compat}(E') \subseteq \mathrm{Compat}(E)$ does not require $\mathrm{Compat}(E')$ to be nonempty; that condition is required by the definition of the determination statuses, so that they are well defined. By contrast, if evidence is removed, or revised in a contradictory way, determination may be lost.

The main result establishes the mathematical conditions that allow us to prove the strict superiority of one alternative over another when both are evaluated by a finite number of minimum and maximum (MIN/MAX) operations, even if some of the values needed to compute them remain unknown, and even when they share leaf variables whose unknown values vary independently over closed, bounded intervals.

Let $L_E(x)$ and $U_E(x)$ be the exact semantic extrema of the value of x:

- The infimum and the supremum over all completions compatible with the evidence E.
- Strict dominance of b over a in all compatible completions holds if and only if $L_E(b) > U_E(a)$.

These extrema are not the bounds computed by a particular algorithm: an implementation typically produces valid but possibly loose bounds. Separation of valid bounds is sufficient for dominance (a classical result), but only separation of the exact extrema is equivalent to it. We prove the converse by constructing a single compatible completion, by threshold clipping, in which b does not exceed a, without requiring disjoint subtrees.

Our proof relies on the established algebra of lattice polynomial functions, and we do not claim originality for the exact characterization. We present a worked example showing that five observed leaves determine a comparison that remains open with only four, with explicit opposite countermodels.

A final section examines, through a conditional countermodel, the separation between computational correctness and moral justification.

A search algorithm returns an action. The fact that it returned it is observable; asserting that it is better than another alternative is a conclusion, and that conclusion requires something that the isolated fact does not provide.

The question guiding this article is:

Under what conditions does the available evidence about a decision problem allow us to justify that one alternative is superior to another?

The question has two complementary aspects:

- The logical aspect examines what an observed selection or an isolated execution can and cannot demonstrate (Section 3).
- The computational aspect examines when partial evidence about a finite MIN/MAX evaluation graph already determines a comparison, so that no completion of the unobserved part can reverse it (Sections 4–5).

1.1 Contributions of the article

The article presents six logical-mathematical contributions that distinguish the observed algorithmic selection, the determination of comparisons by partial evidence, and the attribution of strict preferences under a declared standard.

First. An algorithmic selection does not by itself entail a demonstrated strict preference.

We establish a formal framework that distinguishes three properties: the actual selection of an alternative, the determination of a comparison by the available evidence, and the attribution of a strict preference under a declared standard.

By means of Proposition 2 and the construction of a countermodel, we show that observing the choice of an alternative does not suffice to establish its superiority under that standard. Likewise, Proposition 3 establishes that resolving a comparison is not necessarily equivalent to demonstrating a strict preference. These distinctions set logical limits on the conclusions that can be drawn from the observable behavior of a computational system.

$$\mathcal{K}_{sel} \nvDash \left(Sel_h(b) \Rightarrow DP_h^C(b,a) \right)$$

Proposition 2. Non-entailment between selection and demonstrated preference.

Second. Partial evidence can determine strict superiority.

Definition 2 establishes the semantic conditions that allow us to determine a relation from the available evidence. A comparison can be mathematically determined without knowing all the values involved, provided that it remains true in all compatible completions.

$$E \vDash R \Leftrightarrow \forall \gamma \in \mathrm{Compat}(E) : \gamma \vDash R$$

Definition 2. Semantic determination of a relation.

Third. Separation of exact extrema is necessary and sufficient.

We consider finite MIN/MAX evaluation expressions, including those that share leaf variables. Under assumption H1, according to which the unknown values of distinct variables vary independently over closed, bounded, nonempty real intervals, we prove an exact characterization of strict dominance.

Theorem 1 establishes that the evidence guarantees the strict superiority of (b) over (a) in all compatible completions if and only if the exact lower extremum of (b) exceeds the exact upper extremum of (a).

$$\left[\forall \gamma \in \mathrm{Compat}(E) : V_\gamma(b) > V_\gamma(a) \right] \Leftrightarrow L_E(b) > U_E(a)$$

Theorem 1. Necessary and sufficient characterization of strict dominance.

The sufficiency direction follows directly from the properties of the extrema. Necessity requires showing that, when the extrema are not strictly separated, there exists a compatible completion that prevents us from asserting universal superiority.

Fourth. The characterization admits shared variables.

By means of Lemma 4, which is constructive, we show that, when there is no strict separation of the exact extrema, a single compatible completion can be obtained that refutes strict dominance. This construction respects the identity of shared variables and does not require the compared expressions to have disjoint subtrees.

$$L_E(b) \leq U_E(a) \Rightarrow \exists \gamma \in \text{Compat}(E) : V_\gamma(b) \leq V_\gamma(a)$$

Lemma 4. Construction of a common completion under H1.

The importance of this result is that the necessity of the condition established by Theorem 1 does not depend on assigning incompatible values to a single variable when it appears in different parts of the evaluated structure.

Fifth. A single superior counterexample refutes universal correctness.

Proposition 1 establishes that it suffices to identify one admissible instance in which there exists a legal alternative strictly superior to the selected one in order to refute a claim of universal correctness with respect to the declared domain and objective function.

$$(\exists x \in D, \exists c \in L(x) : V(x, c) > V\,(x, A(x))) \Rightarrow \neg UC(A, D, V)$$

Proposition 1. Refutation of universal correctness by a counterexample.

This result allows us to distinguish the logical refutation of a universal claim from the empirical assessment of the frequency of errors. A single counterexample can invalidate the former without providing, by itself, an estimate of the latter.

Sixth. Consistent extension of the evidence preserves determined relations.

Lemma 2 establishes that, when new evidence restricts the set of compatible completions without altering the semantics, previously determined relations remain determined.

$$(\emptyset \neq \text{Compat}(E') \subseteq \text{Compat}(E)) \wedge (E \vDash R) \Rightarrow (E' \vDash R)$$

Lemma 2. Monotonicity of determination under consistent extensions of the evidence.

This property allows us to distinguish between the preservation of a mathematically justified conclusion and its eventual recognition or certification by a computational procedure.

Verification through a constructive example

We construct a twelve-leaf MIN/MAX structure in which five observations suffice to determine a strict comparison. When any one of them is removed, the remaining evidence admits compatible completions in which the relation under examination holds and others in which it does not.

The twelve-leaf example thus demonstrates, for that particular structure, the difference between a determined and an undetermined comparison, and establishes the inclusion-minimality of the five observations used, without presupposing global minimality.

Epistemic contribution and grounding of RAI

Taken together, these contributions provide a logical-mathematical basis for the epistemic and evidential dimension of Responsible AI (RAI). Their contribution consists in establishing what can legitimately be inferred from a selection, when the evidence allows us to demonstrate a superiority, and what additional guarantees its computational certification requires.

The research thus distinguishes between the output produced by a system, the mathematically justified conclusion, and the human responsibility to examine the criteria and limits of that justification. This distinction allows us to ground a central demand of epistemic responsibility: not to attribute to a computational decision more knowledge, certainty, or justification than the evidence allows us to establish.

1.2 Scope and limits of the results

The results hold within the explicit assumptions of the model: Theorem 1 concerns finite MIN/MAX expressions and compatible completions, admits shared variables, and does not extend, without further proof, to other functions or computational structures. That a comparison is determined does not guarantee that an implementation has computed it correctly, that the chosen alternative is globally optimal, or that the decision is morally justified: that requires additional conditions or premises. The contribution is to formulate the problem of justification by partial evidence, to characterize it within the model, and to establish the logical-epistemic distinctions that allow us to examine the evidential scope of algorithmic decisions.

1.3 The proof of strict dominance requires explicit mathematical assumptions

The mathematical characterization we develop requires specifying what information is regarded as known and what values the not-yet-observed variables may take.

These conditions determine the set of completions compatible with the evidence and, therefore, the scope of the conclusions that can be proved. In particular, we distinguish between the independence of the unknown values and the possibility that one and the same variable appears in different parts of the evaluation graph.

This distinction is essential for establishing Theorem 1 without imposing unnecessary restrictions on the structure of the compared expressions.

H1 — independence of the unknown values.

Each unobserved leaf variable can take any value within a closed, bounded, nonempty real interval. Under H1, the set of completions compatible with the evidence comprises all admissible combinations of those values, while the observed variables remain fixed.

Formally, under H1,

if $K \subseteq \Lambda$ is the set of observed variables and Λ the finite set of leaf variables, then:

$$\mathrm{Compat}(E) = \left\{ \gamma : \Lambda \to \mathbb{R} \;\middle|\; \begin{array}{cc} \gamma(\ell) = e_\ell, & \forall \ell \in K, \\ \gamma(\ell) \in [m_\ell, M_\ell], & \forall \ell \in \Lambda \setminus K \end{array} \right\}.$$

It is assumed that the observed values are admissible and that $\mathrm{Compat}(E) \neq \emptyset$.

Independence refers to distinct unknown variables, not to their occurrences in the graph. If a variable occurs in both compared expressions, all of its occurrences necessarily receive the same value in each completion.

It is thus independence in the domain of admissible values, not a hypothesis of statistical independence.

H2 — disjointness of the unknown variables.

The condition previously considered, which required that the compared expressions share no unknown variables, is not necessary for the strengthened Theorem 1. Lemma 4, included here, allows us to prove the characterization of strict dominance through a single compatible completion, even when shared variables exist.

Consequently, H1 is sufficient for the strengthened result, without imposing H2.

This does not mean that H1 is necessary in all possible models, but rather that it establishes the conditions under which the equivalence has been proved.

The limitations of this characterization in the presence of additional constraints or dependencies among variables are examined in the sections devoted to Theorem 1 and its applications.

1.4 The research applies known mathematical foundations to a problem of justification by partial evidence

The work lies at the intersection of bound-based algorithmic search, the algebraic theory of MIN/MAX functions, and the logic of semantic entailment.

Search and bounds. The use of lower and upper bounds to discard alternatives is classical: Berliner (1979), Winands, Björnsson, and Saito (2008), Cazenave and Saffidine (2010), and Knuth and Moore (1975). It is well known that if a valid lower bound of one alternative exceeds a valid upper bound of another, strict superiority follows; we do not claim that principle or the sufficiency direction of Theorem 1.

MIN/MAX algebra. Finite expressions of minima and maxima are lattice polynomial functions, with known properties of monotonicity and thresholds (Goodstein 1967; Couceiro and Marichal 2010, 2012).

Our contribution is Theorem 1: *strict dominance in all compatible completions is equivalent to separation of the exact extrema, even with shared variables.* The proof is self-contained, but an exhaustive comparison with the existing literature is lacking; for that reason we claim no mathematical priority.

Logic. Semantic entailment is understood in the sense of Tarski (1936). Resolved Comparison (RC) and Demonstrated Preference (DP) come from the prior analysis (Valenzuela Alonso 2026b) and are formulated here through abstract definitions.

Delimitation of the contribution

The research does not propose a new search algorithm or a new algebraic theory of MIN/MAX operations. It formulates a problem of comparative determination under partial evidence, establishes its semantic conditions, and proves a necessary and sufficient characterization within the declared model. Its mathematical validity is examined through proofs; its relation to prior work requires a rigorous bibliographic comparison; and its usefulness for verifying algorithmic decisions would require additional computational validations.

1.5 Organization of the results and reading guide

After fixing the formal framework (compatible completions and determination statuses), the auxiliary results establish the limits of inference from an observed selection. The main theorem characterizes when partial evidence guarantees strict superiority, and a twelve-leaf example

illustrates it. Next, semantic determination is distinguished from computational certification and the conditional applications are examined, among them the audit of UCT and PUCT. The conclusions summarize the proven results and their epistemic significance. A final section, separated from the proofs, examines why computational correctness does not guarantee moral justification.

2 Formal framework

2.1 Compatible completions delimit what partial evidence can prove

We consider a finite, directed, acyclic, rooted evaluation graph (G), whose internal nodes perform MIN or MAX operations and have at least one successor.

Each terminal node is associated with a variable from a finite set Λ.

The same variable may appear at different nodes, including in the expressions corresponding to the compared alternatives.

Each variable $\ell \in \Lambda$ is assigned a closed, bounded, nonempty real interval $I_\ell = [m_\ell, M_\ell]$.

The evidence (E) fixes the values e_ℓ of a subset $K \subseteq \Lambda$.

The model may restrict the admissible assignments. Let Ω be the set of admissible assignments, with $\Omega \subseteq \prod_{\ell\in\Lambda} I_\ell$: assignments $\gamma : \Lambda \to \mathbb{R}$ with $\gamma(\ell) \in I_\ell$ that, in addition, satisfy whatever constraints the model imposes among the variables. The observed values are assumed to be admissible.

A compatible completion is an assignment of Ω that respects the values fixed by the evidence:

$$\mathrm{Compat}(E) = \{\gamma \in \Omega \mid \gamma(\ell) = e_\ell,\ \forall \ell \in K\}$$

Under assumption H1, once the evidence is fixed, the unknown variables can take, independently of one another, any value in their respective closed, bounded, nonempty real intervals; the observed values remain fixed. Consequently, under H1:

$$\mathrm{Compat}(E) = \left\{ \gamma : \Lambda \to \mathbb{R} \;\middle|\; \begin{array}{ll} \gamma(\ell) = e_\ell, & \ell \in K, \\ \gamma(\ell) \in I_\ell, & \ell \in \Lambda \setminus K \end{array} \right\}.$$

Each variable receives a single value per completion, regardless of the number of times it appears in the graph. For a completion γ, the value $V_\gamma(x)$ is obtained by assigning to each terminal node the value of its variable and recursively applying the MIN/MAX operations.

The finiteness and acyclicity of the graph guarantee that this evaluation is well defined.

Definition 1. Exact semantic extrema.

For any node (x), assuming $\mathrm{Compat}(E) \neq \emptyset$, we define:

$$L_E(x) := \inf_{\gamma\in\mathrm{Compat}(E)} V_\gamma(x),$$
$$U_E(x) := \sup_{\gamma\in\mathrm{Compat}(E)} V_\gamma(x).$$

A pair of bounds $\underline{B}_E(x)$ and $\overline{B}_E(x)$ is valid if, for every compatible completion, it satisfies:

$$\underline{B}_E(x) \leq V_\gamma(x) \leq \overline{B}_E(x).$$

Consequently:

$$\underline{B}_E(x) \leq L_E(x) \leq U_E(x) \leq \overline{B}_E(x).$$

The extrema $L_E(x)$ and $U_E(x)$ take into account all compatible completions; they are not statistical estimates. Under H1, both are attained, as shown in the lemma on evaluation at the extrema.

Interpretation of Definition 1: exact compatible bounds

1. Interpretation of the formula: extrema and compatible completions The formula of Definition 1 identifies the two operations that delimit the possible values of a node (x) when the available evidence is partial.

- The infimum ranges over the set Compat(E), over which the lower extremum $L_E(x)$ is determined. This value represents the lower limit of the evaluations $V_\gamma(x)$ obtained by considering all completions compatible with the evidence.
- The supremum ranges over the same set of completions, now used to determine the upper extremum $U_E(x)$. This value represents the upper limit of the possible evaluations of the node.

Both operations consider exactly the same set Compat(E). The difference lies in the extremum being determined: the infimum establishes the lower limit and the supremum the upper limit.

Under assumption H1, these extrema are attainable. Therefore, they correspond respectively to the minimum and maximum value that $V_\gamma(x)$ can take without contradicting the evidence.

The importance of this distinction is that it allows us to delimit mathematically all the values compatible with the available information, without resorting to probabilistic estimates or presupposing values for the unknown variables.

2. Mathematical meaning, scope, and conditions of validity Definition 1, called "Exact Semantic Extrema," fixes the lower and upper limits of the values that an evaluation node can take when only part of the information needed to compute it is known.

To determine these limits, we consider all assignments of values that respect the available evidence.

- The lower extremum is the infimum of the values compatible with that evidence.
- The upper extremum is the supremum of the values compatible with that evidence.

In this way, the definition allows us to delimit the set of possible results without yet knowing the values of all the variables.

This first definition does not constitute a proof. It establishes what both extrema mean mathematically, but it does not itself prove how to compute them, nor does it guarantee that a computer program can compute them correctly.

The existence and computation of the extrema are justified later by the extremal evaluation lemma, under Assumption 1 (H1), which allows the values of the unknown variables to be chosen independently within their admissible intervals.

It is necessary to distinguish the exact mathematical extrema from the bounds that an algorithm can provide. The former take into account all assignments compatible with the evidence. The latter may be approximate and require independent justification to guarantee that they correctly delimit the possible values.

From the standpoint of knowledge, this definition establishes which values remain possible given the available information and the conditions of the model. It does not determine which of those values will occur, does not assign probabilities, and does not compute confidence intervals.

This distinction will be fundamental for Theorem 1, which characterizes strict dominance between two alternatives under partial evidence.

Thus, its proof establishes the necessary and sufficient conditions for one alternative to exceed another in all compatible assignments, using their exact extrema. By contrast, valid bounds that are not exact can provide a sufficient condition for superiority, but not necessarily a necessary condition.

3. Meaning of the symbols and conditions of the model To interpret the definition of the exact extrema correctly, it is necessary to distinguish between the known information, the values that remain unknown, and the different ways of completing that information.

We use the following notation:

Symbol	**Mathematical meaning**
Λ	Set of leaf variables of the evaluation graph.
$K \subseteq \Lambda$	Variables whose value has been observed.
(E)	Available evidence: known values of the observed variables.
Compat(E)	Set of admissible assignments of the model that respect the evidence. Under H1, these are all the assignments that respect the evidence and the intervals.
γ	A specific completion: assigns an admissible value to each unknown variable and keeps the observed values.
$V_\gamma(x)$	Value of node (x) obtained through the MIN/MAX operations under the completion γ.
$L_E(x)$	Exact lower extremum: infimum of the values compatible with the evidence.
$U_E(x)$	Exact upper extremum: supremum of the values compatible with the evidence.
$[m_\ell, M_\ell]$	Interval of admissible values of the variable ℓ.

All values are expressed on the evaluation scale defined for the problem. No specific physical unit is presupposed.

Assumption 1 (H1). Independence of the unknown values

To show how the exact extrema are computed, we use Assumption 1 (H1), defined above.

This assumption states that each unknown variable can take any value in its interval, independently of the values chosen for the other unknown variables. The observations remain fixed.

Consequently, all combinations of admissible values are allowed, without introducing additional constraints among distinct variables.

Independence does not mean that a shared variable can receive different values. If one and the same variable appears several times in the graph, all its occurrences necessarily receive the same value in each completion.

This condition allows us to identify two particular completions: the one in which all unknown variables simultaneously take their minimum values and the one in which they take their maximum values.

These two completions form the basis of the following result.

Lemma 1. Computing exact extrema using extremal assignments

Statement of the problem.

The preceding definition establishes that the exact extrema are obtained by considering all compatible completions. However, it does not yet indicate how to compute them.

The question is whether it is necessary to examine all those completions or whether we can directly identify those that produce the extreme values.

We will show that, under H1, it suffices to evaluate the graph twice: once assigning the minimum values to all unknown variables and once assigning their maximum values.

Statement.

Let (x) be a node of a finite MIN/MAX evaluation graph. We define two compatible completions:

- γ^{-}: assigns to each unknown variable the lower endpoint of its interval.
- γ^{+}: assigns to each unknown variable the upper endpoint of its interval.

In both completions, the observed variables keep their known values.

Then:

$$L_E(x) = V_{\gamma^-}(x), \qquad U_E(x) = V_{\gamma^+}(x).$$

That is, the smallest compatible value of a node is obtained by evaluating it with all unknown variables at their minima; the largest, with all of them at their maxima.

Proof

Step 1. Property of the MIN/MAX operations. The minimum and maximum functions are monotone nondecreasing with respect to each argument.

Increasing any of its arguments, with the others held constant, never decreases the result.

Since the value of each node results from a finite composition of these operations, we can prove by induction on the height of the node that its evaluation is also monotone with respect to the values of the leaf variables.

This property is preserved when a variable appears several times: increasing its value modifies all its occurrences coherently and cannot decrease the result of an expression built solely from MIN/MAX.

Step 2. Comparison of the completions. By construction, for every unknown variable ℓ and every compatible completion γ, we have:

$$m_\ell \leq \gamma(\ell) \leq M_\ell.$$

The observed variables keep the same value in all completions.

Therefore, comparing the assignments variable by variable:

$$\gamma^- \leq \gamma \leq \gamma^+.$$

The inequalities are understood coordinate-wise.

Step 3. Application of monotonicity. Since MIN/MAX evaluation is monotone, the order of the assignments implies the same order between the resulting values:

$$V_{\gamma^-}(x) \leq V_{\gamma}(x) \leq V_{\gamma^+}(x)$$

for every $\gamma \in \text{Compat}(E)$.

Consequently,

no compatible completion can produce a value below $V_{\gamma^-}(x)$ or above $V_{\gamma^+}(x)$.

Step 4. Attainment of the extrema. Assumption H1 guarantees that both γ^- and γ^+ belong to the set of compatible completions, because the endpoints of all the intervals are admissible values and there are no additional constraints among distinct variables.

Therefore, the two preceding bounds are attained.

In accordance with Definition 1 of exact semantic extrema:

$$L_E(x) = V_{\gamma^-}(x), \qquad U_E(x) = V_{\gamma^+}(x).$$

□

Mathematical consequence

The lemma shows that, under H1, it is not necessary to enumerate all compatible completions in order to compute the exact extrema.

Therefore, two evaluations of the graph suffice: one with the lower assignment and one with the upper assignment.

This result is important for the subsequent development, because it allows us to compute the values $L_E(x)$ and $U_E(x)$ that enter Theorem 1, which is devoted to characterizing when the evidence guarantees the strict superiority of one alternative over another.

This lemma does not establish the correctness of a search algorithm, nor does it automatically extend to models that introduce dependencies among unknown variables.

This lemma applies a known fact from interval analysis (Moore 1966): a monotone function attains its extrema at the endpoints of the intervals. Its role here is to provide the exact evaluation of L_E and U_E used by Theorem 1.

Interpretation of Lemma 1: evaluation at the extrema

1. What evaluation at the extrema proves Lemma 1 establishes that, under Assumption 1 (H1), the extreme values of a finite MIN/MAX expression can be obtained without individually examining all the completions compatible with the evidence.

To this end, we consider two assignments: γ^-, which assigns to each unknown variable the lower endpoint of its interval, and γ^+, which assigns it the upper endpoint. In both, the observed values remain fixed and shared variables keep the same value in all their occurrences.

The result establishes the following equalities:

$$L_E(x) = V_{\gamma^-}(x) \quad U_E(x) = V_{\gamma^+}(x)$$

These equalities have a precise meaning. Evaluation under γ^- does not merely provide a valid lower bound, but the smallest value compatible with the evidence. Likewise, evaluation under γ^+ attains the largest compatible value.

The difference is fundamental: we are not working with estimates or approximations of the extrema, but with their exact mathematical values. Under H1, both assignments are compatible and effectively attain the extrema defined in Definition 1.

2. Mathematical and computational scope The consequence is that two evaluations of the MIN/MAX graph suffice to compute its exact semantic extrema, one with the lower assignment and one with the upper. It is not necessary to enumerate the Cartesian product of the admissible intervals.

If the graph contains n nodes and m arcs, and each node is evaluated only once per pass, both extrema can be computed in time:

$$O(n + m)$$

This complexity corresponds to evaluating the graph once the terminal assignments are known. The result relies on the monotonicity of the MIN/MAX operations and on the compatibility of the extremal assignments guaranteed by H1.

The characterization admits shared variables and does not require the compared expressions to have disjoint sets of unknown variables. However, it does not automatically extend to domains with additional constraints among variables, nor to operations other than MIN and MAX, even if they are monotone.

The importance of the lemma for this research is that it provides the exact extrema used later in Theorem 1. Lemma 1 establishes how to compute those extrema; Theorem 1 characterizes when their separation determines a strict superiority between alternatives.

This distinction keeps the computation of the extreme values separate from the mathematical justification of a comparison.

3. Meaning of the symbols and conditions of validity

Symbol	**Meaning**
(x)	Node of the MIN/MAX evaluation graph.
(E)	Available evidence: values of the observed variables.
γ^-	Completion that assigns the admissible minima to the unknown variables.
γ^+	Completion that assigns the admissible maxima to the unknown variables.
$V_{\gamma^-}(x)$	Value of the node obtained under the lower assignment.
$V_{\gamma^+}(x)$	Value of the node obtained under the upper assignment.
$L_E(x)$	Exact lower extremum compatible with the evidence.
$U_E(x)$	Exact upper extremum compatible with the evidence.
H1	Assumption of independence of the unknown values within their admissible intervals.

The values are expressed on the evaluation scale defined for the problem. No specific physical units are presupposed.

Conclusion.

Under H1, the two extremal assignments yield exactly the minimum and the maximum compatible with the evidence. This property is neither an estimate nor an approximation: it is a proven consequence of the MIN/MAX mathematical structure.

2.2 A relation is determined when all compatible completions satisfy it

Suppose that an algorithmic system must choose between two alternatives, (b) and (a).

- The system selects (b), but part of the information needed to evaluate both alternatives remains unknown.
- The selection is a fact we can observe. The superiority of (b) over (a), by contrast, is a claim that we must justify.

The difference raises a question: what would the evidence have to establish for us to assert that (b) is superior to (a), regardless of the values still unknown?

To answer, we consider all compatible completions: assignments of the unknown variables that respect the observations and the constraints of the model.

If the superiority relation holds in all those completions, the evidence suffices to establish it.

If no completion allows it to be sustained, the evidence establishes its falsity. But there is a third possibility:

- that some completions make the relation true and others make it false.

In this last case, the evidence does not demonstrate the superiority, but neither does it necessarily demonstrate its negation.

The distinction does not depend on how many times an algorithm has selected an alternative, nor on the confidence we attribute to its response. It depends on which propositions are true in all the situations still compatible with the available information.

Definition 2. Determination, refutation, and indeterminacy

Let (E) be the available evidence and let Compat (E) be the nonempty set of completions compatible with it.

Consider a proposition (R), whose truth or falsity can be evaluated in each completion.

We define three possible statuses.

A relation is determined when all compatible completions make the proposition true:

$$\text{Status}_E(R) = \text{DETERMINED} \Leftrightarrow \forall \gamma \in \text{Compat}(E) : \gamma \vDash R.$$

The evidence is sufficient to establish (R), because no admissible completion constitutes a counterexample.

A relation is refuted when all compatible completions make the proposition false:

$$\text{Status}_E(R) = \text{REFUTED} \Leftrightarrow \forall \gamma \in \text{Compat}(E) : \gamma \vDash \neg R.$$

In this case, the evidence allows us to establish the negation of (R).

A relation remains undetermined when there exist compatible completions that disagree about its truth:

$$\begin{aligned} &\text{Status}_E(R) = \text{UNDETERMINED} \\ \Leftrightarrow\ &\exists \gamma_1, \gamma_2 \in \text{Compat}(E) : \gamma_1 \vDash R \wedge \gamma_2 \vDash \neg R. \end{aligned}$$

Therefore, indeterminacy does not describe an error of the algorithm or a probability of being mistaken. It identifies a precise property of the evidence: it admits at least one situation in which

the proposition is true and another in which it is false.

Why it is necessary to distinguish these statuses

Consider a minimal example.

Let (b) and (a) be two alternatives whose values depend on a still unknown variable ℓ, with admissible interval ($[-1, 1]$).

Suppose:

$$V_\gamma(b) = \gamma(\ell), \qquad V_\gamma(a) = 0.$$

We want to establish whether (b) is strictly superior to (a):

$$R: \quad V_\gamma(b) > V_\gamma(a).$$

If we do not yet know the value of ℓ, there exist compatible completions with different outcomes. For example, when $\gamma(\ell) = 0.5$, the proposition is true; when $\gamma(\ell) = -0.5$, it is false.

The available evidence does not determine the comparison.

Now suppose that an observation fixes $\ell = 0.5$. All completions compatible with the new evidence satisfy the strict superiority of (b). The comparison becomes determined.

Finally,

if the observation fixes $\ell = 0$, no compatible completion satisfies strict superiority. The comparison is refuted, although this does not mean that (a) is strictly superior to (b): in this case, both alternatives have the same value.

The example shows why the distinction is necessary. Not determining a superiority, determining its falsity, and determining the opposite superiority are different statements. Confusing them leads to attributing to the evidence conclusions that are not justified.

Logical basis of the classification

The three statuses constitute an exhaustive and mutually exclusive classification under classical logic, provided that at least one compatible completion exists.

Indeed, consider the set of completions in which (R) is true.

Only three possibilities arise:

- it comprises all compatible completions;
- it is empty;
- it contains only some of them.

In the first case, the relation is determined.

In the second, it is refuted.

In the third, there exist completions that satisfy (R) and others that satisfy its negation; therefore, it is undetermined.

The classification follows from the definition and from classical bivalent logic. It is not a new theorem about MIN/MAX and does not presuppose Assumption 1 (H1).

That assumption will be needed later to prove a specific characterization of strict dominance, but not to define what it means for a proposition to be determined.

Semantic entailment: what it means for the evidence to demonstrate a relation

The preceding distinction can be expressed through the notion of semantic entailment.

We write:

$$E \models R$$

when (R) is true in all completions compatible with (E).

Therefore:

$$E \models R \Leftrightarrow \text{Status}_E(R) = \text{DETERMINED}.$$

Likewise, $E \models \neg R$ means that the evidence refutes (R).

Now,

that a proposition is not entailed by the evidence does not mean that its negation is:

$$(E \not\models R) \not\Rightarrow (E \models \neg R).$$

This epistemic distinction occupies a fundamental place in the present research.

The absence of sufficient evidence to assert a proposition is not equivalent to having evidence to deny it.

Strictly speaking, we are speaking here of semantic entailment: what is true in all compatible models.

We do not yet claim that a particular computational procedure can correctly recognize that status or produce a certificate of it.

Application to the comparison of alternatives

In our research, the main proposition expresses the strict superiority of one alternative (b) over another (a), under the same objective function:

$$R_{b,a} : \quad V_\gamma(b) > V_\gamma(a).$$

The evidence determines that superiority when:

$$\forall \gamma \in \text{Compat}(E) : V_\gamma(b) > V_\gamma(a).$$

This condition establishes exactly what we want to know: whether the comparison remains true whatever the admissible completion of the unknown values.

However, the definition does not yet provide a procedure for deciding when this condition holds. In principle, the set of completions may contain infinitely many possible assignments.

Here the central mathematical problem of the article arises.

Is it possible to determine strict superiority by examining only the exact extrema of the alternatives, without having to consider individually all compatible completions?

Theorem 1 answers affirmatively for finite MIN/MAX expressions under Assumption 1.

We will show below that, in that model, strict superiority is determined exactly when the lower extremum of (b) exceeds the upper extremum of (a):

$$\big(\forall \gamma \in \mathrm{Compat}(E) : V_\gamma(b) > V_\gamma(a)\big) \Leftrightarrow L_E(b) > U_E(a).$$

Definition 2 therefore fixes what it means for a comparison to be determined by the evidence. Theorem 1 characterizes mathematically when that determination can be established through the exact extrema.

Epistemic scope: a common foundation for human reasoning and algorithmic evaluation

This classification allows us to raise an epistemic question that goes beyond the mere description of results:

Under what conditions can a person recognize as justified a conclusion obtained by a machine?

An algorithmic response is an observable outcome.

- However, the existence of that outcome does not on its own establish its truth, nor does it show that it is sufficiently supported by the available information.
- Justifying a claim requires identifying its premises and checking whether the evidence excludes the completions that contradict it.

Definition 2 provides a logical framework for making that distinction.

- When a proposition is true in all completions compatible with the evidence, its semantic determination is established.
- When it is false in all of them, the evidence allows us to refute it.
- When there exist compatible completions that disagree, the proposition remains undetermined.

These three statuses are not three degrees of confidence or three truth values. Each completion evaluates the proposition as true or false. The classification expresses something different: what the evidence as a whole allows us to establish about that proposition.

Therefore, determination does not measure the probability that the proposition is true. Nor is it equivalent to the number of favorable observations. It refers to a relation of logical consequence between the evidence and the proposition under examination.

This difference is especially important in the interaction between people and artificial intelligence systems.

A system can produce a response without having sufficient information to justify it.

It can also correctly identify that a conclusion is determined within a formal model, provided that the procedure used is adequate and its correctness has been established.

In both cases, the person examining the result needs to distinguish what the system has produced from what can actually be demonstrated.

Here a possible common foundation between human reasoning and algorithmic evaluation appears: **the requirement that a claim not exceed what its premises and evidence allow us to conclude.**

It is not required that a person and a machine reason in the same way, but that the conclusions under evaluation be open to examination by explicit and verifiable criteria.

- The person can ask why an alternative is considered superior.
- The system can provide the evidence, the comparison criterion, and, where appropriate, a certificate whose validity can be checked.

The decisive issue is not that both produce the same answer, but that it be possible to examine the reasons that authorize us to sustain it.

The difference between *indeterminacy* and *refutation* is particularly important in this context.

If the available evidence does not allow us to prove that an alternative (b) exceeds another (a), we cannot automatically conclude that (b) is inferior.

Nor can we regard the two alternatives as equivalent.

All that has been established is that the evidence does not yet determine the relation under examination.

Formally:

$$(E \nvDash R) \nRightarrow (E \vDash \neg R).$$

This distinction prevents the transformation of an insufficiency of knowledge into a claim of falsity.

It also introduces a shared epistemic responsibility:

the system should not present as demonstrated what its evidence does not establish, and the person should not attribute to the result a certainty that has not been justified.

Nevertheless, the separation between a semantic property and its computational recognition must be maintained.

That a relation is determined in all compatible completions does not automatically show that a particular program has correctly identified that status. To make that attribution, it is necessary to establish the correspondence between the operations of the program and the mathematical conditions of the model.

This requirement is an essential part of our research:

to distinguish the output produced, the mathematically determined conclusion, and the attribution of a demonstrated preference.

Thus, Theorem 1 will provide, under the assumptions declared for MIN/MAX evaluations, a necessary and sufficient characterization of the determination of strict superiority.

Its contribution does not consist in replacing human judgment, but in establishing what comparative relation is guaranteed by the evidence within that model.

Practical decision-making raises an additional question.

- A person may have to act even when the evidence does not determine which alternative will produce the best result.
- The indeterminacy of a comparison does not necessarily eliminate the reasons for deciding, just as a mathematically determined comparison does not on its own establish the moral legitimacy of the corresponding action.

For this reason, **truth must not be confused with decision**, **nor mathematical justification with the responsibility to act.**

The epistemic contribution consists in making explicit the limits of each of these questions.

The mathematical model allows us to establish certain relations between evidence and conclusion; computational evaluation requires procedures whose correctness can be verified; and human

judgment can incorporate reasons, obligations, and principles that are not derived exclusively from that comparison.

In this sense, **the research provides a formal basis for articulating the encounter between human reasoning and algorithmic evaluation**: not through an identity between the two, but through the possibility of subjecting their claims to common conditions of justification.

- A machine can produce a choice.
- A proof allows us to establish what conclusion the evidence supports.
- A person can examine that justification and assess what consequences should follow from it.

The distinction is essential:

selection is a fact; demonstrated preference is a conclusion subject to conditions of evidence; and the responsibility to act requires a judgment that is not exhausted by calculation.

Thus, the value of semantic determination does not lie solely in establishing when a comparison is true in all compatible completions. It also lies in delimiting, explicitly and in a way open to examination, what we know, what we can prove, and what remains unknown.

The meeting point between human reasoning and algorithmic evaluation is not the coincidence of their decisions, but the *possibility* of examining what *evidence* justifies each conclusion and recognizing the *limits* of what can be *demonstrated*.

Interpretation of Definition 2: semantic status

1. What the equivalence expresses The equivalence that defines determination (Definition 2) has two components.

First component — Status of the relation.

- The expression $\text{Status}_E(R) = \text{DETERMINED}$ indicates that the available evidence is sufficient to establish the proposition (R) as entailed by the evidence.

This status does not represent a choice of the algorithm or an assessment of confidence. It expresses that the truth of (R) is fixed by the conditions of the model and the evidence considered.

Second component — Condition that justifies the determination.

The expression

$$\forall \gamma \in \text{Compat}(E) : \gamma \vDash R$$

establishes the necessary and sufficient mathematical condition: every completion compatible with the evidence satisfies the proposition.

The universal quantifier $\forall$ is essential. It is not enough to find one completion that confirms (R), or even several. Determination requires the relation to be true in all admissible completions.

The equivalence between the two components explains the result: the evidence determines a proposition precisely when there is no compatible completion that could refute it.

This condition does not guarantee that a computer program has correctly recognized the determination.

It establishes what must hold mathematically for such recognition to be valid.

2. Meaning of the symbols and conditions of interpretation

Symbol	Meaning
(E)	Available evidence: information observed and fixed in the model.
(R)	Proposition whose truth is examined.
$\text{Status}_E(R)$	Semantic status of (R) with respect to the evidence (E).
$\text{Compat}(E)$	Nonempty set of completions compatible with the evidence.
γ	A complete assignment of admissible values.
$\forall\gamma$	Universal quantification: the condition must hold for all compatible completions.
$\gamma \vDash R$	The proposition (R) is true under the completion γ.
$\Leftrightarrow$	Logical equivalence: both conditions hold in exactly the same cases.

The definition presupposes a classical interpretation of truth and a nonempty set of compatible completions. By itself, it does not require the independence assumption H1.

2.3 Selecting, resolving a comparison, and demonstrating a preference are different properties

The selection of an alternative is an observable fact; its superiority over another requires additional justification.

To distinguish the two questions, we examine whether the available evidence allows us to resolve a comparison and, if so, what relation it establishes.

Consider an execution identified by the history (h), two distinct alternatives (b) and (a), and a declared evaluation standard (C).

The expression $Selected_h(b)$ indicates that the algorithm selected (b), without thereby attributing any superiority to it.

Let $\mathcal{W}(h)$ be the set of possible comparison witnesses. Each witness κ is an evidence record subject to two conditions:

- *admissibility*, expressed by $Adm_C(\kappa; h, b, a)$, which requires its correct linkage to the history, the alternatives, and the standard;
- *resolution*, expressed by $Resolved_C(\kappa; h, b, a)$, which requires that the evidence satisfy the conditions necessary to establish a comparison relation under (C).

The witness is not a person, nor the algorithm, nor the verifier: it is a concrete evidence record that allows us to justify a comparison between alternatives.

We define a Resolved Comparison (RC) when there exists a witness that satisfies both conditions:

$$RC_h^C(b, a) \Leftrightarrow \exists\kappa \in \mathcal{W}(h) : Adm_C(\kappa; h, b, a) \\ \wedge Resolved_C(\kappa; h, b, a).$$

Resolving a comparison does not imply that its result favors (b): it may also establish a tie or the superiority of (a).

Therefore, we define a demonstrated strict preference (Demonstrated Preference, DP) only when the same admissible witness that resolves the comparison establishes the strict superiority of (b) over (a):

$$DP_h^C(b,a) \Leftrightarrow \exists \kappa \in \mathcal{W}(h) : Adm_C(\kappa; h, b, a) \\ \wedge Resolved_C(\kappa; h, b, a) \\ \wedge Rel_C(\kappa) = b \succ_C a.$$

Here, $Rel_C(\kappa)$ identifies the relation attested by the witness, and $\succ_C$ represents strict superiority under the standard (C).

It follows from these definitions that every demonstrated strict preference implies a Resolved Comparison, but not conversely: a witness that attests a tie resolves the comparison without demonstrating a strict preference.

These definitions are abstract.

Attributing a preference to the decision actually selected requires verifying, in addition, the corresponding operational conditions, including the linkage to the observed selection.

The fundamental distinction is that selecting, resolving a comparison, and demonstrating a strict superiority are different properties, which must not be confused or attributed without the required evidence.

2.4 Operational attribution of preference to the selected alternative

The preceding definitions define when a comparison is resolved (RC) and when the evidence attests a strict superiority between two alternatives (DP). However, these properties do not by themselves allow us to assert that the demonstrated preference corresponds to the decision actually taken by the algorithm.

Demonstrating that an alternative is superior under a given criterion is not equivalent to demonstrating that the algorithm selected it, nor that the evidence used can be validly attributed to that execution.

To make this attribution, the following must be verified jointly: the observed selection, the resolution of the comparison, the direction of its result, and the operational conditions governing the admissibility, scope, and provenance of the witness.

To distinguish them from the abstract comparative definitions, we introduce the following formalization.

- Let (h) be the history of an execution;
- (b), the alternative whose selection is examined, and (a), a distinct alternative;
- (C), the comparison standard, and (e), a concrete evidence record.

We define the operational witness of demonstrated preference by:

$$DPWith_h^{C,op}(b,a;e) \Leftrightarrow Selected_h(b) \\ \wedge RC_h^{C,op}(b,a;e) \\ \wedge Rel_C(e) = b \succ_C a \\ \wedge Sat\left(\Gamma_C^{op}; h, b, a, e\right).$$

The expression establishes four necessary and jointly sufficient conditions for the attribution defined by this scheme:

Effective selection: the history attests that the algorithm selected alternative (b).

Resolved comparison:

- the record (e) satisfies the conditions of operational resolution of the comparison between (b) and (a).

Strict superiority attested:

- the recorded relation strictly favors (b) under the standard (C).

Operational validity of the witness:

- the record satisfies the additional conditions collected in Γ_C^{op}, including whatever checks of scope, linkage, and provenance are required.

The logical conjunction $\wedge$ requires that the four conditions hold for the same history, the same alternatives, the same standard, and the same evidence record. Satisfying a single condition does not suffice to justify the attribution.

It should be noted that the predicate $DPWith_h^{C,op}(b, a; e)$ refers to a concrete witness.

To assert that there exists an operational attribution of demonstrated preference, we require at least one valid witness:

$$DP_h^{C,op}(b, a) \Leftrightarrow \exists e \in \mathcal{E}_h^C(b, a) : DPWith_h^{C,op}(b, a; e).$$

Here, $\mathcal{E}_h^C(b, a)$ represents the set of candidate records associated with the execution, the pair of alternatives, and the standard under consideration.

This distinction allows us to identify situations in which a favorable comparison exists but not an operationally attributable preference.

For example, a comparison could be correctly resolved and favor (b), while the history shows that the algorithm selected another alternative.

In that case, the comparative superiority could hold, but the attribution of demonstrated preference to the selection of (b) would not be justified.

Similarly, a favorable result does not suffice if the evidence record violates a mandatory condition of provenance or linkage.

The formulation therefore distinguishes three facts that require different checks: which alternative the algorithm selected, what relation the evidence allows us to demonstrate, and whether that relation can be validly attributed to the observed execution.

These definitions are abstract: they do not show that a specific implementation of the RC–DP verifier, documented in the second article (version 0.25), satisfies them. That correspondence requires checking that all its conditions of admissibility, resolution, scope, linkage, and provenance are faithfully represented; it is not presupposed by having written the formula.

Consistency of the conclusions obtained from different witnesses

A comparison may be supported by more than one evidence record.

An additional question then arises:

what happens if two records admitted for the same comparison lead to different conclusions?

The preceding definitions use an existence condition: it suffices to find one admissible witness that resolves the comparison.

However, that condition does not guarantee that other witnesses, equally admitted, establish the same result.

Consider two witnesses, κ_1 and κ_2, corresponding to the same execution (h), two alternatives (b) and (a), and a comparison standard (C).

Suppose that both satisfy the conditions of admissibility and resolution, but that the first establishes the strict superiority of (b), while the second establishes the strict superiority of (a).

In that case,

both comparisons can be regarded as resolved under the existential definitions, even though their conclusions are incompatible.

This shows that the existence of a witness supporting strict superiority does not by itself exclude the existence of another witness that contradicts it.

To guarantee a non-contradictory comparative conclusion, it is necessary to examine agreement among witnesses:

all admissible witnesses that resolve the same comparison, under the same standard and for the same history, must establish the identical relation.

The condition requires equality of results, not identity of the witnesses.

Two records may contain different evidence and be equally valid, provided that they establish the same comparison relation.

This agreement does not follow from the preceding definitions.

None of the preceding definitions requires all admitted witnesses to coincide.

Agreement is therefore an additional condition that must be justified through the properties of the evaluation standard or through the validation rules for witnesses.

Thus, mere coherence of the input data does not necessarily guarantee that property if the procedures used to interpret them can produce incompatible results.

We say that a standard C is coherent for a comparison (h, b, a) when all of its admissible witnesses that resolve the comparison establish the same comparison relation. If C is coherent, no admissible witness that resolves the comparison can contradict a demonstrated strict preference under that standard. Coherence is a universal property over those witnesses and does not by itself guarantee that any exists: the existence of a demonstrated preference remains governed by DP.

Coherence is guaranteed when all admitted witnesses are semantically sound with respect to one and the same consistent evidence E, one and the same comparison, and one and the same criterion, and the possible comparison relations (superiority of b, superiority of a, or a tie) are mutually exclusive: two incompatible relations cannot be true at once in one and the same compatible completion.

By contrast, a standard that does not guarantee such soundness may admit conflicting witnesses. In that case, it is not justified to attribute a conclusively demonstrated preference until the incompatibility is resolved. This contradiction does not show that the preference is false: it establishes only that the admitted records do not justify an unambiguous conclusion.

The subsequent propositions are proved from the existential definitions of Resolved Comparison and Demonstrated Preference. They do not presuppose universal agreement among witnesses, unless expressly indicated.

The existence of a comparative justification and the consistency of all admitted justifications are distinct properties: neither should be inferred automatically from the other.

Interpretation of demonstrated strict preference (DP)

Inference scheme We can express this reasoning in the following scheme:

$$\frac{\begin{array}{c}\kappa \in \mathcal{W}(h), \qquad Adm_C(\kappa; h, b, a), \\ Resolved_C(\kappa; h, b, a), \\ Rel_C(\kappa) = (b \succ_C a)\end{array}}{DP_h^C(b, a)}$$

Reading of the scheme. If we find a witness κ among the candidate records associated with the execution, check that it is admissible under the standard (C), verify that it resolves the comparison, and establish that its result states the strict superiority of (b), then we can assert that there exists a demonstrated strict preference ("Demonstrated Preference," DP) of (b) over (a), under that standard.

The expressions above the line represent the required conditions. Below appears the conclusion that those conditions allow us to establish.

The scheme shows the direct application of the definition, not an independent theorem. The defining equivalence also permits the inverse reasoning: to assert that there exists a demonstrated strict preference requires that there exists at least one witness jointly satisfying those conditions.

What a demonstrated strict preference does and does not assert A demonstrated strict preference is relative to the declared standard. An alternative may be superior under a criterion of speed and not under another of accuracy, and strict superiority under C does not automatically imply superiority under another evaluation function V, whose correspondence with C would have to be justified separately.

Nor should it be confused with the actual selection: an algorithm may select b without having a witness that demonstrates its superiority over a, and there may be evidence favorable to b without the algorithm having selected it.

The absence of a valid witness does not show that b is inferior: it means that strict superiority has not been attested by the available witnesses. The lack of demonstration and the demonstration of the opposite relation must be kept separate.

This requirement offers an evidential basis for Responsible AI (RAI). The resulting thesis places in human oversight the definition of the standard, the examination of the reasons, and the responsibility to justify its application.

Meaning of the terms used To interpret the formula we must distinguish the elements involved in it:

Expression	**Meaning**
$\mathcal{W}(h)$	Set of candidate records that can be examined as witnesses of a comparison associated with the history (h).
κ	Evidence record examined to justify the conclusion.
$Adm_C(\kappa; h, b, a)$	The record meets the admissibility conditions of the standard (C) for that execution and those alternatives.
$Resolved_C(\kappa; h, b, a)$	The record satisfies the conditions necessary to resolve the comparison.
$Rel_C(\kappa)$	Comparison relation established by the witness.
$b \succ_C a$	Strict superiority of (b) over (a) under the standard (C).
$DP_h^C(b, a)$	Existence of at least one admissible witness that resolves the comparison and demonstrates that strict superiority.

The conceptual result is the following:

a demonstrated strict preference does not consist in attributing an intention to the algorithm or in accepting its choice as proof.

It consists in establishing a comparison relation through evidence that satisfies the expressly declared conditions.

2.5 Determination, certification, and attribution

Four statements must not be confused, with $R_{b,a}$ the proposition $V_\gamma(b) > V_\gamma(a)$:

(i) the fact $Selected_h(b)$;

(ii) the semantic determination $\text{Status}_E\left(R_{b,a}\right) = \text{DETERMINED}$;

(iii) the acceptance of a certificate by a verifier, which would support (b) only under an explicit hypothesis of verifier correctness, not studied here;

(iv) the attribution $DP_h^C(b,a)$ with respect to a standard.

Theorem 1 concerns exclusively (ii).

To relate (ii) with (iv), we expressly adopt the following modeling convention.

Definition 3. Bound-Separation Standard C_V.

For an objective function V and evidence E, a witness κ_E for (b,a) consists of E and its exact bounds (L_E, U_E) for V. It records $b \succ_{C_V} a$ if and only if $L_E(b) > U_E(a)$. Admissibility and resolution are regarded as properties of E (consistency and common objective function V).

This is a stipulated semantic standard, not a consequence of the original operational definitions.

We define the canonical bound record $\kappa_E = (E, V, L_E, U_E, b, a)$ for every nonempty compatible evidence with well-defined exact extrema.

If admissibility under this stipulated standard includes the existence and correct linkage of κ_E, and if resolution is established by checking the exact bounds, then, under H1, the following holds:

$$\begin{gathered}\kappa_E \text{ is admitted and certifies } b \succ_{C_V} a \\ \Leftrightarrow \text{Status}_E\left(R_{b,a}\right) = \text{DETERMINED}.\end{gathered}$$

It is a mathematical statement about the constructed record and about Theorem 1, not a proof that any implemented verifier has generated, accepted, or checked that record.

Without the premises of existence and admissibility of the canonical record, Theorem 1 provides only semantic determination and not the existence of an operational certificate of DP. No similar equivalence is claimed for other standards C.

Interpretation of Definition 3: from Theorem 1 to the standard of demonstrated preference

The condition of Definition 3 derives from the result proved in Theorem 1, but it serves a different function. The theorem shows when strict superiority holds in all compatible completions; the standard defines how we will recognize and attest that superiority through exact bounds.

To use it as a criterion of demonstrated preference, we require that the comparison witness contain the corresponding exact bounds, be correctly linked to the alternatives and to the evidence examined, and satisfy the admissibility conditions of the standard.

When these conditions are guaranteed and all witnesses admitted for this comparison use the same evidence (E), the demonstrated strict preference ("Demonstrated Preference," DP) under this standard is characterized by:

$$\begin{gathered}DP_h^{C_V}(b,a) \Leftrightarrow L_E(b) > U_E(a) \\ \Leftrightarrow \forall \gamma \in \text{Compat}(E) : V_\gamma(b) > V_\gamma(a).\end{gathered}$$

- The first equivalence depends on the definition and on the admissibility conditions of the standard.

- The second follows from Theorem 1.

Meaning and scope of the result The consequence is precise:

under the established conditions, we can attest a strict superiority without knowing all the individual values, provided that the available evidence determines the exact bounds and these satisfy the proven inequality.

If that inequality does not hold, the evidence does not guarantee the universal superiority of (b). This does not necessarily mean that (a) is superior: the completion that blocks the conclusion may establish a tie.

The result likewise does not show that an algorithm selected (b), nor that a computer program has correctly computed and verified the bounds. Those claims require additional checks.

The central mathematical contribution of this section is therefore the equivalence proved by Theorem 1. The standard defined from it allows us to transfer that result to an explicit criterion of attestation, without confusing the mathematical truth of a comparison with the correctness of an implementation.

In summary:

Theorem 1 shows when the evidence guarantees a strict superiority; the bound-separation standard establishes how to attest it. The existence of an algorithmic choice remains a different fact from the demonstration of its superiority.

A decision can be a verifiable fact without the superiority of the chosen alternative being a demonstrated conclusion. The critical task is to determine what the evidence authorizes us to assert and what remains outside that scope.

3 Auxiliary results

3.1 Universal correctness of an algorithm

Consider a deterministic algorithm (A), whose version, rules, and parameters remain fixed, and a set (D) of admissible instances.

- For each instance $x \in D$, let $(L(x))$ be the finite, nonempty set of legal actions. We will assume that the algorithm selects an action $A(x) \in L(x)$.
- Let $(V(x, a))$ be the value assigned to a legal action (a) in instance (x), according to a fixed, declared objective function. These values are established by an evaluation procedure independent of the algorithm under examination.

What does it mean to assert that the algorithm always selects an optimal action? It means that, for each instance of the declared domain, the selected action attains the maximum value among all legal actions.

We define this property as universal correctness ("Universal Correctness"):

$$UC(A, D, V) \Leftrightarrow \forall x \in D : A(x) \in \arg\max_{a \in L(x)} V(x, a).$$

The expression arg max represents the set of actions attaining the maximum value. If several actions share that maximum, any of them satisfies the optimality condition.

The formula therefore asserts that the algorithm selects an optimal action in all instances of the examined domain.

This is a defined property, not a claim that the algorithm under study actually satisfies it.

Proposition 1. A single counterexample suffices to refute universal correctness

Suppose that we find an admissible instance $x_0 \in D$ in which the algorithm selects $(A(x_0))$.

- If there exists another legal action $a' \in L(x_0)$ whose value is strictly superior to that of the selected action, then the algorithm does not satisfy universal correctness in the declared domain.

  Formally:

$$\exists x_0 \in D,\ \exists a' \in L(x_0) : \qquad V(x_0, a') > V(x_0, A(x_0)) \Rightarrow \neg UC(A, D, V).$$

Proof. By hypothesis, there exists an instance (x_0) and a legal action (a') whose value strictly exceeds that of the action chosen by the algorithm.

Consequently, $(A(x_0))$ does not belong to the set of actions that maximize (V) in that instance:

$$A(x_0) \notin \arg\max_{a \in L(x_0)} V(x_0, a).$$

The definition of universal correctness requires that this membership hold for every instance of the domain (D).

Since we have identified an instance in which it does not hold, the universal claim is refuted.

$$\neg UC(A, D, V).$$

□

Observation. The converse also holds: if the universal condition fails at (x_0), the selected action does not maximize (V); since $(L(x_0))$ is finite and nonempty, some legal action attains a strictly greater value.

Meaning of the result

The proposition establishes that a single, correctly attested case suffices to refute the claim that the algorithm always selects an optimal action.

The refutation does not depend on the frequency of failures and does not allow it to be estimated: a single valid counterexample invalidates the universal claim.

Its scope, however, is delimited.

It refers exclusively to the configuration of the algorithm (A), the domain (D), and the objective function (V) used in the evaluation.

In addition, the superiority of the alternative action must be attested by correct values under that same objective function. A higher heuristic score, by itself, does not show that another action is strategically superior.

The proposition establishes the evidence necessary to refute the universal correctness of a given configuration; it does not on its own establish the failure of a specific algorithm.

3.2 An algorithmic selection does not by itself entail a demonstrated strict preference

Proposition 2. Selection does not entail demonstrated preference.

Let K_{sel} be a nonempty class of structures that interpret *Selected*, W, *Adm*, *Resolved*, and *Rel* as in Section 2.3, and that contains, for the pair examined, a structure with a recorded selection of b but without admissible comparison witnesses (for example, $W(h) = \emptyset$).

Then, for that pair $b \neq a$:

$$Selected_h(b) \not\models_{K_{sel}} DP_h^C(b, a).$$

Proof.

By the explicit hypothesis of the existence of a model, take $M \in K_{sel}$ with $Selected_h(b)$ true and such that no $\kappa \in W(h)$ satisfies $Adm_C(\kappa; h, b, a)$ (for example, $W(h) = \emptyset$). The existential quantifier of the definition of DP (Section 2.3) then lacks a witness, so that $DP_h^C(b, a)$ is false in M. The premise is true and the conclusion false in the same model; the semantic entailment is thereby refuted. □

The following results specify what can be concluded from an algorithmic execution and what evidence suffices to justify or refute a claim. They are logical consequences of the established definitions; they do not replace the constructive proof of Theorem 1, which must retain its independent and complete development.

3.3 Resolving a comparison is not necessarily equivalent to demonstrating a strict preference

A Resolved Comparison (RC) requires an admissible witness that establishes a comparison relation; demonstrated preference requires, in addition, that this relation strictly favor (b).

Proposition 3. Resolved comparison does not necessarily entail demonstrated preference.

The first implication follows from the definitions; the converse fails, in general, in classes of models that allow comparisons resolved by a tie:

$$DP_h^C(b, a) \Rightarrow RC_h^C(b, a), \qquad RC_h^C(b, a) \not\Rightarrow DP_h^C(b, a).$$

Proof. The witness that attests strict preference also satisfies the conditions of a resolved comparison. To refute the converse, it suffices to take a model with a single admissible witness that resolves the comparison and whose result is $b \sim_C a$ (tie): there is resolution, but no strict preference. ▫

The existence of a witness supporting strict superiority likewise does not guarantee, by itself, that other admitted witnesses do not contradict it: that agreement requires additional conditions of the standard.

3.4 Consistent extension of the evidence preserves determined relations

Lemma 2. Monotonicity of determination under consistent evidence extension.

Let $E \subseteq E'$ be two evidence sets interpreted under the same semantics and with $\mathrm{Compat}(E') \neq \emptyset$, where $\mathrm{Compat}(E)$ is the set of completions compatible with (E).

For any proposition (R):

$$\mathrm{Compat}(E') \subseteq \mathrm{Compat}(E), \qquad E \vDash R \Rightarrow E' \vDash R.$$

Proof. Every completion that satisfies the extended evidence (E') also satisfies (E).

If (R) was true in all completions compatible with (E), it remains true in its subset compatible with (E'). ▫

This property concerns semantic determination:

it does not show that a program automatically generates new certificates or that the statuses of its operational records are maintained.

3.5 Example CE09: the objective function matters

In the synthetic case CE09 described in the research material (61 nodes and six actions at the root), a local heuristic (H) favors (a_2) over (a_0), while minimax evaluation (V) reverses that comparison:

$$H(a_2) = 0.95 > -0.50 = H(a_0), \qquad V(a_2) = 0.40 < 0.90 = V(a_0).$$

If the implementation selects (a_2), the legal action (a_0) constitutes a counterexample of superiority under the minimax objective, with a value difference of (0.50) between the two.

A favorable heuristic score does not demonstrate optimality under another objective function.

External verification of these data and of any claim of independent checking requires consulting the corresponding experimental archive.

4 Main theorem: characterization of strict superiority under partial evidence

A favorable comparison in a single evaluation does not demonstrate universal superiority when part of the information remains unknown.

Other compatible completions could reverse the result.

Consider two alternatives b and a, evaluated by finite expressions built exclusively from MIN and MAX operations.

Let E be the available evidence and $\mathrm{Compat}(E)$ the nonempty set of completions compatible with it.

- For each completion γ, the expressions $V_\gamma(b)$ and $V_\gamma(a)$ represent the values of both alternatives under the same assignment of the unknown variables.

The central question is the following:

What condition allows us to assert that b is strictly superior to a in all completions compatible with the evidence?

The proof proceeds in two stages.

- First we establish a sufficient condition based on valid bounds.
- Then we show that, when the exact extrema are used and the domain-independence assumption holds, that condition is also necessary.

4.1 Sufficient separation of valid bounds

A valid lower bound $\underline{V}_E(x)$ and a valid upper bound $\overline{V}_E(x)$ must satisfy, for every compatible completion γ,

$$\underline{V}_E(x) \leq V_\gamma(x) \leq \overline{V}_E(x).$$

These bounds may be conservative: they need not coincide with the smallest and largest values actually attainable.

Lemma 3. Sufficient separation of valid bounds

If

$$\underline{V}_E(b) > \overline{V}_E(a),$$

then b is strictly superior to a in all compatible completions:

$$\underline{V}_E(b) > \overline{V}_E(a) \Rightarrow \forall \gamma \in \text{Compat}(E) : V_\gamma(b) > V_\gamma(a).$$

Proof.

For any compatible completion γ,

$$V_\gamma(b) \geq \underline{V}_E(b) > \overline{V}_E(a) \geq V_\gamma(a).$$

Therefore,

$$V_\gamma(b) > V_\gamma(a)$$

for every $\gamma \in \text{Compat}(E)$.

□

An example illustrates Lemma 3. Suppose that b takes a value in [0.9, 1] and a in [0, 0.5]. Neither of the two values is known, but $b > a$ in all cases, because the worst case of b (0.9) exceeds the best case of a (0.5).

The lemma expresses the classical principle of bound separation: if even a lower bound of b exceeds an upper bound of a, no compatible completion can reverse the comparison.

However,

the lack of separation between approximate bounds does not allow us to conclude that the superiority is undetermined. The bounds may be too wide.

To obtain a necessary and sufficient condition we must use the exact semantic extrema of Definition 1, L_E and U_E.

4.2 The difficulty of the converse direction

Suppose now that the exact extrema are not strictly separated:

$$L_E(b) \leq U_E(a).$$

It might seem sufficient to combine the worst possible value of b with the best possible value of a. But that reasoning is not sufficient.

The lower extremum of b may be attained under one completion, while the upper extremum of a may be attained under another. If both expressions share variables, we cannot assign two different values to one and the same variable in order to realize both extrema simultaneously.

Therefore,

to refute universal strict superiority we need to construct a single compatible completion satisfying:

$$V_\gamma(b) \leq V_\gamma(a).$$

The following lemma constructs that completion.

4.3 A common completion refutes strict dominance when the extrema are not separated

We adopt Assumption 1:

Each unknown variable can take any value in its closed, bounded, nonempty real interval, independently of the other unknown variables. The observed variables keep their values.

The compared expressions may share variables.

When a variable appears several times, all its occurrences receive a single value in each completion.

Lemma 4. Existence of a common completion

Let $V_\gamma(b)$ and $V_\gamma(a)$ be two finite MIN/MAX expressions.

Under Assumption 1, if

$$L_E(b) \leq U_E(a),$$

then there exists a compatible completion γ_t such that

$$V_{\gamma_t}(b) \leq V_{\gamma_t}(a).$$

Proof

Take as the threshold

$$t = L_E(b).$$

For each unknown variable ℓ, with admissible interval

$$I_\ell = [m_\ell, M_\ell]\,,$$

we define:

$$\gamma_t(\ell) = \max\{m_\ell,\, \min\{t, M_\ell\}\}\,.$$

This threshold-clipping operation assigns to ℓ the value of the interval closest to the threshold t:

if $t < m_\ell$, it assigns m_ℓ;

if $m_\ell \le t \le M_\ell$, it assigns t;

if $t > M_\ell$, it assigns M_ℓ.

The observed variables keep the values fixed by E. Assumption 1 guarantees that the resulting assignment belongs to Compat(E).

Let γ^- be the completion that assigns to all unknown variables their lower endpoints and γ^+ the one that assigns them their upper endpoints.

By the lemma on evaluation at the extrema,

$$V_{\gamma^-}(b) = L_E(b) = t, \qquad V_{\gamma^+}(a) = U_E(a).$$

The proof uses the following property of MIN/MAX expressions.

For any real threshold s,

$$\max(u, v) > s \Leftrightarrow (u > s) \vee (v > s),$$

and

$$\min(u, v) > s \Leftrightarrow (u > s) \wedge (v > s).$$

By structural induction on the expression, the proposition

$$V_\gamma(x) > s$$

is obtained by replacing the MAX operations with disjunctions and the MIN operations with conjunctions of the corresponding leaf predicates.

Thus, this property remains valid when a variable is repeated.

First inequality:

$V_{\gamma_t}(b) \le t$

For each unknown variable ℓ,

$$\gamma_t(\ell) > t \Leftrightarrow m_\ell > t.$$

Therefore,

with respect to the threshold t, the leaves under γ_t and under γ^- have the same truth values.

The observed leaves coincide by definition.

Consequently,

$$V_{\gamma_t}(b) > t \Leftrightarrow V_{\gamma^-}(b) > t.$$

But

$$V_{\gamma^-}(b) = t,$$

so that $V_{\gamma^-}(b) > t$ is false. Therefore,

$$V_{\gamma_t}(b) \leq t.$$

Second inequality:

$$V_{\gamma_t}(a) \geq t$$

Consider any real number $s < t$.

If an unknown variable exceeds s under the upper assignment γ^+, it also exceeds it under γ_t. Indeed,

$$M_\ell > s \Rightarrow \gamma_t(\ell) > s.$$

The monotonicity of the conjunctions and disjunctions that represent the MIN/MAX thresholds implies:

$$V_{\gamma^+}(a) > s \Rightarrow V_{\gamma_t}(a) > s.$$

Since

$$V_{\gamma^+}(a) = U_E(a) \geq t > s,$$

it holds that

$$V_{\gamma_t}(a) > s$$

for every $s < t$. Consequently,

$$V_{\gamma_t}(a) \geq t.$$

Combining both inequalities:

$$V_{\gamma_t}(b) \leq t \leq V_{\gamma_t}(a).$$

Therefore,

$$V_{\gamma_t}(b) \leq V_{\gamma_t}(a).$$

The constructed completion assigns a single value to each variable, even when that variable appears in both compared expressions.

□

Clarification of the construction. Two properties of γ_t, valid for all variables at once, support the proof. With respect to the threshold t, γ_t coincides with γ^- as to which variables strictly exceed it: $\gamma_t(\ell) > t \Leftrightarrow m_\ell > t$. And, for every $s < t$, every variable that exceeds s under γ^+ also exceeds it under γ_t: $M_\ell > s \Rightarrow \gamma_t(\ell) > s$. Since MAX and MIN are equivalent to disjunctions and conjunctions of these predicates, and each variable receives a single value even if it appears in both expressions, the first property implies that b exceeds t under γ_t only if it exceeds it under γ^-, which is false; and the second implies that a exceeds s under γ_t whenever it exceeds it under γ^+. Finally, since γ_t is a compatible completion, $V_{\gamma_t}(b) \geq L_E(b) = t$; together with the first inequality this yields $V_{\gamma_t}(b) = t$, and therefore $V_{\gamma_t}(b) = t \leq V_{\gamma_t}(a)$.

The lemma does not necessarily guarantee that a is strictly superior to b. The constructed completion may produce a tie. This suffices to refute the claim that b is strictly superior in all compatible completions.

4.4 Theorem 1. Necessary and sufficient characterization of strict dominance

After the preceding reflections and proofs, we can answer the central question of the article.

Theorem 1.

Let b and a be two alternatives evaluated by finite MIN/MAX expressions. Suppose that:

the evidence E is consistent;

$\text{Compat}(E) \neq \emptyset$;

each unknown variable belongs to a closed, bounded, nonempty real interval;

Assumption 1 of domain independence holds.

The expressions may share variables.

Then:

$$\forall \gamma \in \text{Compat}(E) : \; V_\gamma(b) > V_\gamma(a) \quad \Leftrightarrow \quad L_E(b) > U_E(a).$$

Interpretation of Theorem 1:

The evidence guarantees the strict superiority of b over a if and only if the exact lower extremum of b exceeds the exact upper extremum of a.

Proof

Sufficiency. Suppose that

$$L_E(b) > U_E(a).$$

For every compatible completion γ,

$$V_\gamma(b) \geq L_E(b) > U_E(a) \geq V_\gamma(a).$$

Therefore,

$$V_\gamma(b) > V_\gamma(a)$$

in all compatible completions.

This direction does not need Assumption 1.

It requires only that the exact extrema be well defined and that the set of compatible completions be nonempty.

Necessity. We now argue by contraposition.

Suppose that

$$L_E(b) \leq U_E(a).$$

By Lemma 4,

there exists a compatible completion γ_t such that

$$V_{\gamma_t}(b) \leq V_{\gamma_t}(a).$$

Consequently,

- it is not the case that b is strictly superior to a in all compatible completions.

By contraposition, if:

$$V_\gamma(b) > V_\gamma(a)$$

for every $\gamma \in \mathrm{Compat}(E)$, then necessarily

$$L_E(b) > U_E(a).$$

Both directions of the equivalence are thus established.

□

4.5 What the theorem actually establishes

The sufficiency direction follows immediately from the exact bounds.

- If even the worst compatible value of b exceeds the best compatible value of a, no completion can reverse the comparison.

The necessity direction requires an additional construction.

If the exact extrema are not separated, Lemma 4 constructs a single admissible completion in which strict superiority fails.

This construction remains valid when the alternatives share variables.

Therefore,

strict separation of the exact extrema is not merely a sufficient criterion.

Under the declared assumptions, it is a necessary and sufficient condition.

The theorem characterizes a semantic property:

which relation is true in all completions compatible with the evidence. It does not show that a computer program has correctly computed the extrema, has recognized the relation, or has issued a valid certificate.

4.6 Why the independence assumption matters

The necessity direction cannot be extended without further argument to domains with dependencies among variables.

Consider two variables related by:

$$x = y + 1, \qquad y \in [0,\ 1],$$

and two alternatives with values

$$V(b) = x, \qquad V(a) = y.$$

In every compatible assignment,

$$V(b) = V(a) + 1 > V(a).$$

However,

$$L_E(b) = 1, \qquad U_E(a) = 1,$$

so that there is no strict separation:

$$L_E(b) \not> U_E(a).$$

This example does not contradict Theorem 1.

The set of admissible assignments is restricted by $x = y + 1$ and is not the full Cartesian product of the intervals of x and y.

This example shows that Assumption 1 is relevant for the necessity direction.

The sufficiency direction remains valid without it.

4.7 The characterization of strict dominance admits shared variables

The compared alternatives may share variables.

Each shared variable receives a single value in each completion, and Lemma 4 uses precisely a common assignment for all its occurrences.

For this reason,

it is not necessary to require the two expressions to have disjoint unknown variables.

The result has been proved for finite expressions built from MIN and MAX over ordered real values.

It does not automatically extend to:

- evidence that imposes additional dependencies among variables;
- non-monotone operators;
- arithmetic functions other than MIN/MAX, even monotone ones, such as addition;
- infinite models without additional conditions of compactness or attainability.

Observation. The equivalence of Theorem 1 does not extend to every monotone function with shared variables. Let $x \in [0,\ 1]$, $f(x) = x + 1$, and $g(x) = x$: $f(x) > g(x)$ holds for

every x, but $\inf f = 1 = \sup g$, so there is no strict separation of extrema. Lemma 4 does not use monotonicity alone, but rather that MIN and MAX are equivalent to conjunctions and disjunctions of threshold predicates over the leaves, with the same threshold s for all of them; addition does not have that property. In the example, $f(x) > s$ is equivalent to $x > s - 1$, whereas $g(x) > s$ is equivalent to $x > s$: the two expressions require different thresholds on the same variable. The common assignment of Lemma 4 therefore cannot preserve the threshold predicates of both at once, and no value of x achieves $f(x) \leq g(x)$.

MIN/MAX expressions belong to the theory of lattice polynomial functions. The monotonicity and the threshold identities used in the proof are known properties of that algebraic structure.

The proof presented is self-contained and establishes the result within the declared model.

No claim of mathematical priority is made over equivalent formulations that may exist in the literature. A claim of originality would require a specific, theorem-by-theorem comparison with the works of Goodstein, Couceiro, and Marichal. The purpose of this analysis is its applicability to Responsible AI (RAI).

4.8 Corollary 1. Unique optimality among all legal alternatives

Theorem 1 compares two alternatives.

To show that one alternative is the unique optimum, we must compare it with all rivals:

- Let $\mathcal{A}(s)$ be the finite, nonempty set of legal actions of a maximization state s, and let $b^\star \in \mathcal{A}(s)$.

  Suppose that, for every rival alternative $a \in \mathcal{A}(s) \setminus \{b^\star\}$,

$$L_E(b^\star) > U_E(a).$$

  Then:

$$\forall \gamma \in \mathrm{Compat}(E) : \arg\max_{x \in \mathcal{A}(s)} V_\gamma(x) = \{b^\star\}.$$

Proof

Fix a compatible completion γ and a rival alternative $a \neq b^\star$.

By the separation of extrema,

$$V_\gamma(b^\star) \geq L_E(b^\star) > U_E(a) \geq V_\gamma(a).$$

Therefore, $b^\star$ strictly exceeds every rival alternative.

Consequently, it is the unique maximizer.

Since the reasoning holds for any compatible completion,

$$\arg\max_{x \in \mathcal{A}(s)} V_\gamma(x) = \{b^\star\}$$

for every $\gamma \in \mathrm{Compat}(E)$.

□

This corollary uses only the sufficiency direction of the separation of extrema and therefore does not need Assumption 1.

The conceptual difference is precise:

- exceeding one alternative is a pairwise comparative property;
- being the unique optimum requires exceeding all rival alternatives.

If the inequalities were not strict,

$$L_E(b^\star) \geq U_E(a)$$

for every rival a, we could guarantee that $b^\star$ belongs to the set of maximizers in all completions, but not that it is its only element.

The sufficiency of the corollary does not require H1.

Under H1 and for the MIN/MAX expressions considered, Theorem 1 also proves the converse direction for each rival.

Therefore, under those assumptions the global characterization is obtained:

$$\left[\forall \gamma \in \mathrm{Compat}(E) : \arg\max_{x \in \mathcal{A}(s)} V_\gamma(x) = \{b^\star\}\right] \Leftrightarrow \left[\forall a \in \mathcal{A}(s) \setminus \{b^\star\} : L_E(b^\star) > U_E(a)\right].$$

The epistemic difference is precise:

a demonstrated comparison against one rival does not authorize proclaiming global optimality if other legal alternatives remain uncompared under the same objective.

Choice and freedom of decision raise philosophical questions that the optimality criterion, by itself, does not resolve.

4.9 Logical scope and meaning of Theorem 1

The sufficiency direction establishes that one alternative exceeds another in every completion when even its compatible minimum exceeds the compatible maximum of the rival. This consequence is valid with any nonempty set of completions for which the extrema are well defined:

it requires neither a product domain nor disjointness of variables.

The necessity direction contains the substantive mathematical step:

when there is no strict separation of the exact extrema, the threshold-clipping construction provides one and the same admissible completion that prevents strict superiority.

The negative conclusion does not consist merely in a detector not having found a proof; there exists a mathematical countermodel.

It may be a tie, so the absence of dominance of (b) does not imply strict dominance of (a).

The elimination of H2 constitutes an improvement in scope over the historical formulation: a variable may occur in both expressions without invalidating the equivalence.

Assumption 1, or the product domain, is sufficient for the construction and is not claimed to be the weakest possible condition.

The result likewise does not assert that any implementation has produced, verified, or preserved a certificate. These are additional operational obligations.

The domain assumptions are not a secondary formality: they delimit the set of countermodels that the proof can construct.

H1 allows the admissible values of distinct variables to be combined freely, but does not authorize violating additional constraints that link some variables with others. If the problem imposes correlations, the threshold-clipping completion might not belong to the permitted domain.

By contrast,

sharing a variable does not constitute an additional constraint among distinct variables. It means that two positions refer to one and the same datum and must respect its identity.

This distinction explains why the extended theorem admits expressions with shared information, whereas it does not extend without further argument to correlated assignment domains.

4.10 Admissibility of shared variables

Consider

$$b(x) = \max\{x, 0.7\}, \qquad a(x) = x,$$

with

$$x \in [0,\ 1].$$

The variable is shared and

$$L(b) = 0.7 \leq U(a) = 1.$$

With (x=0.7), we obtain

$$b(x) = a(x).$$

The common completion exhibits the failure of universal strict superiority. This example illustrates the construction; it does not replace the general proof.

The following example does not attempt to measure speed, performance, or hit rate.

Its function is to show, through verifiable values, how partial evidence can suffice to determine a strict relation.

It also illustrates the difference between losing a positive proof and demonstrating that the opposite relation is true.

5 Five observations suffice to determine the comparison in the twelve-leaf MIN/MAX example

Figure 1 represents a MAX root with two compared alternatives, (a_5) and (a_0).

Each alternative leads to a MIN node with three responses (r_0, r_1, r_2), each evaluated by a MAX node with two leaves. All leaves belong to $([-1,\ 1])$.

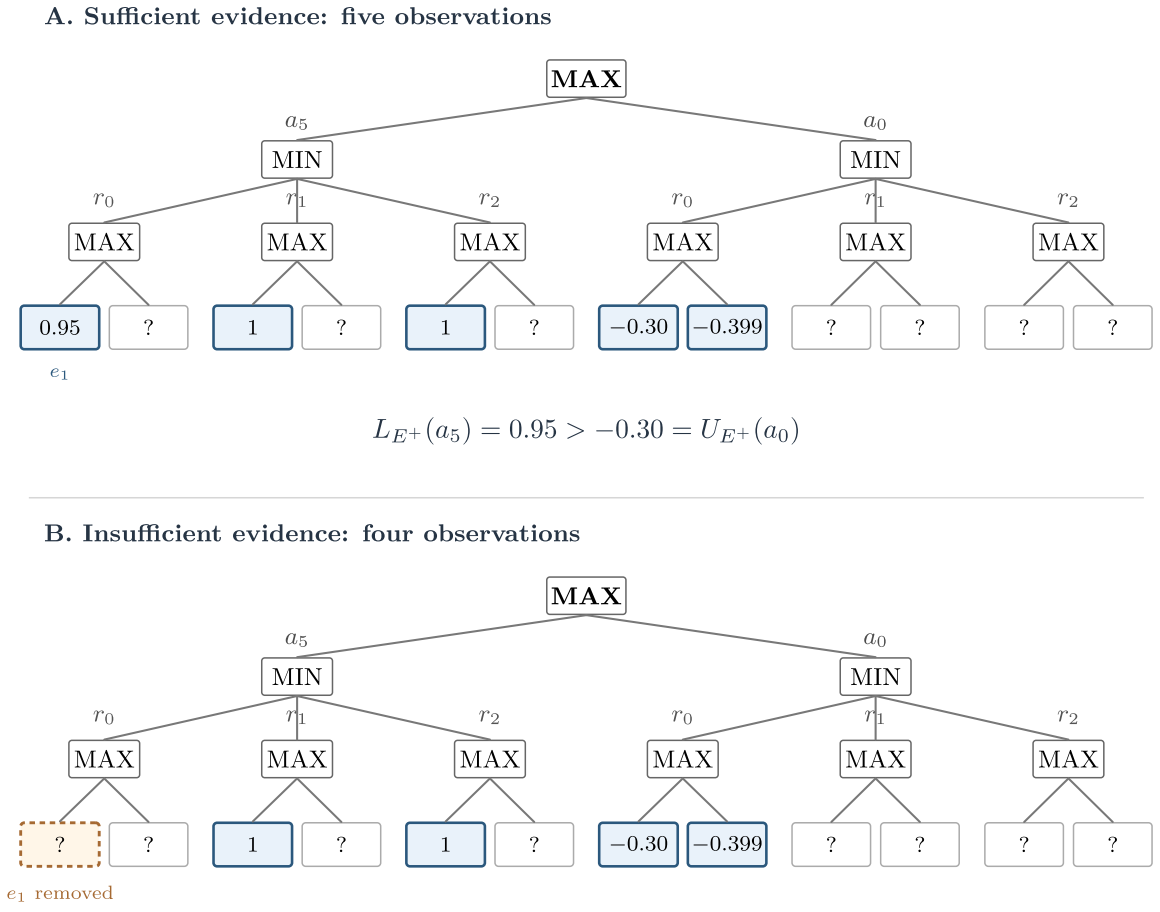


Figure 1. Twelve-leaf MIN/MAX tree. A: five observations suffice to demonstrate the strict superiority of a_5 over a_0. B: when the observation $e_1 = a_5 : r_0 : 1 = 0.95$ is removed, the remaining evidence admits completions with opposite comparative outcomes. Shaded leaves are the observations of E^+; question marks denote unobserved values, with domain $[-1,\ 1]$; the dashed outline marks the leaf that is removed to obtain E^-.

The tree contains two alternatives, each with three branches whose value depends on the maximum of two leaves, then combined by a minimum. This yields twelve leaves in total.

The initial evidence fixes five values and leaves seven unobserved, all the unknowns lying within the indicated admissible interval.

What is computed is the smallest value still possible for the alternative under examination and the largest value still possible for its rival.

In this particular instance, each of the twelve terminal positions represents a distinct semantic variable.

The seven unobserved variables have domain ($[-1,\ 1]$), while the five observed ones are fixed at their declared values.

We denote each leaf by $a : r : k$, where a is the alternative (a_5 or a_0), r the branch (r_0, r_1, or r_2), and $k \in \{1,\ 2\}$ the position within the MAX node of that branch ($k = 1$ is the left leaf and $k = 2$ the right leaf in Figure 1). Thus, the observed leaf of value (0.95) is $a_5 : r_0 : 1$.

This particularization allows values to be assigned freely to distinct unknown leaves. In the general Theorem 1, by contrast, a variable may be repeated and all its occurrences would have to keep the same value.

5.1 Strict determination under five observations

The evidence (E^+) fixes five leaves:

$$a_5 : r_0 : 1 = 0.95,\ a_5 : r_1 : 1 = 1,\ a_5 : r_2 : 1 = 1, a_0 : r_0 : 1 = -0.30,\ a_0 : r_0 : 2 = -0.399.$$

The five observations fixed in the preceding development are the following: for (a_5), the first leaf of each of its three branches, $a_5 : r_0 : 1$, $a_5 : r_1 : 1$, and $a_5 : r_2 : 1$, takes the values (0.95), (1), and (1), respectively.

For (a_0), the two leaves of its first branch, $a_0 : r_0 : 1$ and $a_0 : r_0 : 2$, take the values (-0.30) and (-0.399). The remaining leaves remain unknown in ($[-1, 1]$).

These observations suffice to fix the exact lower extremum of (a_5) at (0.95) and the exact upper extremum of (a_0) at (-0.30).

The reason is structural:

- In (a_5), each MAX branch retains a value at least equal to the observed one, and the top-level MIN cannot fall below (0.95); that extremum is attainable.
- In (a_0), the first MAX branch is fixed at (-0.30) and the top-level MIN can never exceed it; that extremum is also attainable.
- Since (0.95) is strictly greater than (-0.30), every compatible completion maintains the superiority of (a_5). We have not needed to determine the remaining seven values.

For (a_5), the sets of compatible values of its three MAX children are, respectively,

$$[0.95,\ 1], \qquad 1, \qquad 1.$$

The MIN node takes the smallest, and its exact lower extremum is

$$L_{E^+}(a_5) = 0.95.$$

For (a_0), the first MAX is fixed at

$$\max(-0.30, -0.399) = -0.30.$$

Since the top node is MIN, its value can never exceed (-0.30), and that extremum is attained.

Thus,

$$L_{E^+}(a_5) = 0.95 > -0.30 = U_{E^+}(a_0).$$

The inequality demonstrates a strict separation between the two extrema, not a comparison between estimated scores.

By Theorem 1, (a_5) strictly exceeds (a_0) in all compatible completions.

Note that it is not necessary to know the remaining seven leaves: the five observed values suffice to exclude any completion that equals or reverses the comparison.

The proof of sufficiency does not exhaust what this instance teaches.

We remove the observation of value (0.95) from (a_5) and keep the other four. The first MAX branch of (a_5) then ceases to have a positive lower bound fixed by that leaf: its two leaves can take negative values.

Hence, the lower extremum of (a_5) drops to (-1), while the exact upper extremum of (a_0) remains at (-0.30).

This loss of separation does not prove that (a_0) is always superior. It proves that the reduced evidence no longer excludes all completions unfavorable to (a_5).

To establish indeterminacy we must exhibit, in addition, a favorable and an unfavorable completion, both compatible with the remaining four observations.

5.2 Indeterminacy under the removal of one observation

We remove:

$$a_5 : r_0 : 1 = 0.95$$

and call (E^-) the remaining evidence.

Now the first MAX child of (a_5) can take any value from (-1) to (1); the other two continue to take the value (1).

Therefore,

$$L_{E^-}(a_5) = -1.$$

In (a_0) there remains

$$U_{E^-}(a_0) = -0.30.$$

The exact extrema no longer satisfy the strict separation

$$L_{E^-}(a_5) > U_{E^-}(a_0)$$

necessary to certify that superiority.

To show indeterminacy, and not merely the absence of positive certification, we construct two compatible completions.

Completion favorable to (a_5).

We assign (1) to the two leaves of the branch (r_0) of (a_5), that is, $a_5 : r_0 : 1 = a_5 : r_0 : 2 = 1$, and (-1) to the two leaves of the branch (r_1) of (a_0), that is, $a_0 : r_1 : 1 = a_0 : r_1 : 2 = -1$.

We obtain

$$V(a_5) = 1 > -1 = V(a_0).$$

Completion unfavorable to (a_5).

We assign (-1) to the two leaves of the branch (r_0) of (a_5), that is, $a_5 : r_0 : 1 = a_5 : r_0 : 2 = -1$, and (1) to the four unknown leaves of (a_0), that is, those of its branches (r_1) and (r_2): $a_0 : r_1 : 1 = a_0 : r_1 : 2 = a_0 : r_2 : 1 = a_0 : r_2 : 2 = 1$.

We obtain

$$V(a_5) = -1 < -0.30 = V(a_0).$$

The values not specified in both completions may be chosen arbitrarily within their intervals, since they do not affect the inequality.

Both cases respect the four observations.

The relation $(V(a_5) > V(a_0))$ is therefore undetermined under (E^-): there exists at least one completion that satisfies it and another that satisfies its negation.

This conclusion is stronger than stating that no certificate has been found; it demonstrates two opposite outcomes within the admissible domain.

The comparison is not universally refuted by the fact of losing its positive proof.

The freedom to choose the remaining values can be justified branch by branch:

- In the favorable completion, the three MAX branches of (a_5) are fixed at (1), so that its MIN equals (1); one branch of (a_0) is fixed at (-1), so that its MIN equals (-1).

None of the remaining variables can alter these values, since all are bounded between (-1) and (1).

The minimality check can be repeated by removing the five observations one at a time.

- If any one of the three corresponding to (a_5) is removed, at least one branch can drop to (-1) and the global minimum loses its guarantee.
- If either of the two in the first branch of (a_0) is removed, that branch can reach (1) and the upper limit that prevented the reversal disappears.

5.3 Relative minimality of the evidence certificate

- Removing any of the three observations of (a_5) causes its lower extremum to drop to (-1).
- Removing either of the two observations of (a_0) allows its upper extremum to rise to (1).

In both cases the strict separation disappears.

The set (E^+) is thus inclusion-minimal for this certificate: none of its five observations can be removed while preserving the demonstrative inequality.

This does not prove that five is the smallest possible number among all alternative certificates, nor that the certificate is unique.

Explicit check of the five removals.

For each (i) between (1) and (5) we consider

$$E_i = E^+ \setminus \{e_i\},$$

where $(e_1 = a_5 : r_0 : 1)$, $(e_2 = a_5 : r_1 : 1)$, and $(e_3 = a_5 : r_2 : 1)$ are the observations of the branches (r_0), (r_1), and (r_2) of (a_5), while $(e_4 = a_0 : r_0 : 1)$ and $(e_5 = a_0 : r_0 : 2)$ are the two observed leaves of (r_0) in (a_0).

In all five cases a favorable completion exists:

we assign (1) to every still-unknown leaf of (a_5) and (-1) to the two leaves of (r_1) in (a_0), that is, $a_0 : r_1 : 1 = a_0 : r_1 : 2 = -1$.

Then $(V(a_5))$ is (1), and $(V(a_0) = -1)$.

To remove (e_1, e_2) or (e_3), we additionally construct an unfavorable completion by assigning (-1) to both leaves of the branch of (a_5) whose observation was removed, that is, $a_5 : r_j : 1 =$

$a_5 : r_j : 2 = -1$, where (j) is that branch ($j = 0$, $j = 1$, or $j = 2$ according to whether e_1, e_2, or e_3 is removed), and (1) to all the unknown leaves of (a_0).

We obtain

$$V(a_5) = -1, \qquad V(a_0) = -0.30.$$

To remove (e_4) or (e_5), we assign (1) to the released leaf of (r_0) in (a_0), that is, $a_0 : r_0 : 1$ or $a_0 : r_0 : 2$ respectively, and to all the remaining unknown leaves of (a_0).

In (a_5) we assign (-1) to the unknown leaf of (r_0), that is, $a_5 : r_0 : 2 = -1$, and admissible values to the others.

We obtain

$$V(a_5) = 0.95, \qquad V(a_0) = 1.$$

All the assignments respect the four retained observations.

Thus, each reduced evidence set admits a favorable and an unfavorable completion: the comparison is undetermined in all five cases and does not merely fail to satisfy the specific separation certificate.

This shows inclusion-minimality within (E^+), not global minimality of cardinality among arbitrary alternative sets of observations.

The example shows a precise epistemic property:

a comparison can be determined without all the individual values being so.

It also shows that the loss of an observation can reopen contradictory completions.

It is a verifiable synthetic instance, not a statistical estimate of the behavior of an algorithm.

Up to this point we have studied a semantic property: what follows necessarily from a set of observations under certain mathematical assumptions.

This truth does not depend on a program detecting it.

However, when we want to attribute a demonstrated preference to a particular decision, a distinct question arises:

what conditions allow us to trust the record, the witness, and the procedure that recognize that truth?

This distinction takes up the problem left open by the preceding article.

The fact of selecting an alternative does not by itself constitute a demonstrated strict preference.

This research provides a positive mathematical characterization for a certain class of comparisons, but it remains necessary to distinguish the existence of that guarantee from its effective certification by an implementation.

6 Semantic determination does not by itself guarantee valid computational certification

Theorem 1 establishes a semantic entailment relation.

It does not assert that a computer system can always recognize that entailment, nor that an acceptance label suffices to justify it.

Three statements are thus distinguished:

- The proposition is true in all compatible completions.
- A verifier has accepted a record.
- That record constitutes, under a declared standard, an admissible witness of strict preference.

The three may coincide, but their relations require explicit premises.

Two different moments may exist along a sequence of evidence:

the first at which the available information semantically determines a comparison, and the first at which a sound verifier recognizes that determination through an admitted certificate.

The difference between the two is conceptual, even if in a particular execution the moments coincide.

Thus, neither a necessarily positive delay nor a performance advantage should be inferred from it.

6.1 Determination and recognition along a sequence of evidence

Let:

$$E_0 \subseteq E_1 \subseteq \cdots$$

be a sequence of consistent evidence sets under a fixed semantics.

We define t_{sem} as the first index at which

$$E_t \models R$$

and t_{cert} as the first index at which a sound procedure accepts a certificate of (R) using exclusively E_t.

When both exist, it holds that

$$t_{sem} \leq t_{cert}.$$

where (R) is the comparative proposition, t_{sem} identifies the first evidence state that determines it, and t_{cert} the first state at which a sound procedure certifies it.

The inequality is a consequence of soundness: a procedure that accepts only conclusions entailed by its data cannot get ahead of semantic determination.

It is not shown that a strict delay exists, nor is a computational advantage claimed.

The claim presupposes that both indices exist, that the evidence sets form a sequence of consistent extensions, that the semantic assumptions and the comparison domain remain fixed, and that acceptance uses exclusively the evidence available at each index.

Under these conditions,

- Extending the evidence restricts or preserves the set of compatible completions without retrospectively modifying the meaning of the variables.
- Soundness is the condition that allows us to pass justifiably from a computational acceptance to a valid conclusion within the declared domain.

Put differently:

if the verifier accepts a proposition accompanied by evidence and a certificate, that acceptance must imply that the proposition is entailed by the evidence.

Presenting a certificate is not equivalent to showing that the program in charge of checking it is sound.

6.2 Soundness and operational attribution

Let $Ver(E, R, \rho)$ be a verifier that receives evidence (E), a proposition (R), and a certificate ρ.

The soundness property over a declared domain $\mathcal{D}$ requires that any acceptance imply a valid semantic entailment.

In particular,

$$Sound_{\mathcal{D}}(Ver) \wedge [Ver(E, R, \rho) = accepts] \qquad \Rightarrow E \vDash R, \qquad (E, R) \in \mathcal{D}.$$

where ρ is the proof object presented, $Sound_{\mathcal{D}}(Ver)$ is the soundness property of the verifier with respect to the declared domain, and $E \vDash R$ expresses that (R) is entailed by the evidence.

This implication is correct by the definition of soundness; it is not a proof that a particular version of the software satisfies that property.

The connection among Resolved Comparison (RC), demonstrated strict preference (DP), and the possible discrepancy among witnesses requires distinguishing their functions.

RC indicates that there exists an admissible witness capable of resolving a relation; DP requires specifically that the attested relation strictly favor the examined alternative.

Agreement among different witnesses is an additional question that these definitions do not guarantee by themselves.

None of these conditions follows from the mere observation of the selection, nor does it replace the proof of the verifier's soundness.

To additionally attribute demonstrated strict preference to a selected alternative, it is not enough to check the semantic inequality.

The standard must specify which witness is accepted, how it is linked to the history and to the two alternatives, which objective function is used, how the bounds are attested, and how potentially contradictory witnesses are handled.

With the evidence (E) and the objective function (V) fixed, we introduce the semantic standard of separation of exact extrema, represented by C_V, through the definition:

$$b \succ_{C_V} a \Leftrightarrow L_E(b) > U_E(a).$$

The evidence (E) remains fixed in the definition of this standard.

Therefore, although the abbreviation used in the formula highlights the objective function (V), the comparison relation also depends on (E) and on the set Compat(E).

It should not be interpreted as a preference independent of the evidence, the domain, or the state evaluated.

This clarification preserves the notation used in the preceding definitions and avoids attributing to it a greater scope than stipulated.

The equivalence in this expression stipulates the criterion of the standard; it does not, by itself, constitute a new theorem.

Theorem 1 independently establishes that, under H1 and MIN/MAX, that criterion coincides with strict superiority in all compatible completions.

The existence of a relation defined by C_V does not ensure that an implementation has produced a valid witness.

To relate it to demonstrated strict preference $DP_h^{C_V}(b, a)$, it is necessary to establish the existence of a canonical record with correct extrema, its admissibility, the resolution under the standard, and, when the preference is to be attributed to a decision, the fact that (b) was actually selected.

Only under those additional premises could the operational correspondence be established; no such correspondence is claimed for specific versions of a verifier.

This separation prevents two unjustified inferences: identifying the positive output of a program with the truth of the proposition it purports to certify, and converting a preference relative to a heuristic objective into strategic superiority under a different objective.

The main mathematical conclusion admits an affirmative formulation: incomplete evidence can attain sufficient deductive force to determine a comparison.

This result is obtained under an identifiable set of assumptions and with a constructive proof that avoids simultaneously assigning incompatible values to shared variables.

Delimiting the scope does not diminish the importance of the theorem. It avoids confusing a property proved in a precise mathematical class with a generalization about any search, any evaluation function, or any certification program.

The validity of the proof is preserved precisely because its presuppositions remain explicit.

7 Discussion of results and delimitations

- The necessary and sufficient characterization requires preserving its domain assumptions: finite MIN/MAX expressions, consistent evidence, and a product set of admissible intervals for the unknown variables.
- The sufficiency of valid bounds has a wider scope; the necessity proved depends on the common threshold completion.
- The proof does not automatically carry over to models with joint constraints, to other evaluation operations, or to conservative bounds confused with exact semantic extrema.
- The twelve-leaf example is not a performance trial: it is an exact mathematical instance that admits verification by MIN/MAX propagation and by two explicit completions.
- The minimality of its five observations is inclusion-minimality with respect to the chosen certificate, not uniqueness or a global minimum in the number or cost of observations.
- Evidence reconstructed retrospectively should likewise not be confused with the information the implementation used when deciding.
- The characterization relies on known algebraic properties of lattice polynomials; no mathematical priority is proclaimed with respect to the whole literature.
- Likewise, the article does not establish the soundness, completeness, efficiency, or regulatory compliance of an implementation.
- These reservations are conditions for a correct interpretation of the proofs, not denials of their validity in the declared model.

The applied relevance of the characterization can be understood by following a sequence of responsibilities.

- First, it is declared which relation is to be justified and under which objective function.
- Then it is identified which evidence is compatible with the model and which extrema can be attested.
- Finally, if a computational certification is intended, it is checked whether the procedure that examines the certificates satisfies the required guarantees.

Each step adds obligations that are not obtained for free from the previous one.

8 Conditional applications: formal verification, search audit, and responsible AI

- Theorem 1 can serve as a specification for a verifier because it provides a checkable mathematical condition on the relation between evidence and conclusion.
- The specification expresses what a certificate should attest; it does not show that an existing computer system correctly implements that criterion.
- The separation between specification and implementation protects precisely the distinction between mathematical result and operational recognition.

8.1 The theorem as a specification for formal verification

Formal verification studies whether an implementation satisfies a mathematical specification with respect to its inputs, states, and outputs.

Theorem 1 provides an exact semantic specification for MIN/MAX comparisons within the declared domain: a claim of universal strict superiority must correspond to the separation of exact extrema.

This allows us to define proof obligations for a certificate checker, but does not show that an existing program meets them.

To transfer a computational acceptance to semantic entailment, the following are required at a minimum: the validity of the bounds or the correct computation of the extrema, the identity of the objective function and the scale, the preservation of shared variables, the correspondence of the witness with the history and the alternatives, the fulfillment of H1 when the necessity direction is used, and an explicit policy on typing, provenance, and consistency of witnesses.

The soundness property expressed in the section “Soundness and Operational Attribution” is a verifiable premise about the program and its domain, not an automatic consequence of the acceptance of a file.

There is, in addition, a difference between verifier soundness and verifier completeness.

- Soundness requires that every accepted certificate support a semantically valid conclusion.
- Completeness would require that every determined relation in the relevant class could be recognized through the admitted certificates.

Neither of these two properties is attributed here to a specific software version.

UCT and PUCT illustrate why this separation is necessary.

Their selection rules serve to guide exploration, but a high exploration score is not thereby a demonstration of optimal value.

An audit must declare the external objective against which it compares the alternatives and the specific configuration of the execution.

Only then is it legitimate to interpret a counterexample as a refutation of a universal claim formulated over that domain.

8.2 Application to the audit of UCT and PUCT

Tree search using upper confidence bounds, known as UCT (“Upper Confidence Bounds Applied to Trees”; Kocsis and Szepesvári 2006), and its variant PUCT, which incorporates prior probabilities (Rosin 2011) and is used, with modifications, in AlphaZero (Silver et al. 2017, 2018),

are exploration and selection methods that should not be identified without qualification with an exact MIN/MAX evaluator.

Their exploration scores or heuristic estimates are likewise not necessarily valid semantic bounds on the reference value.

However, the decisions produced by those methods can indeed be audited using a declared external value criterion.

When randomness is involved, an auditable execution must be identified by a complete configuration

$$z = (x, B, \theta, \omega, \pi),$$

where (x) is the instance, (B) the simulation budget, θ the parameters of the procedure, ω the seed or random realization, and π the action-presentation order.

Once these components are fixed,

the execution is regarded as determined for the purposes of the countermodel:

- If it selects (b) and a legal action (a) is attested with

$$V(a) > V(b),$$

universal correctness is refuted with respect to any domain of configurations that contains (z), in accordance with Proposition 1.

The conclusion is not a claim of failure in all executions of UCT or PUCT.

In the experimental records on refutation by a superior witness, selections of value (0.65) against alternatives of value (0.90) with budget six are reported, for a synthetic benchmark and specific configurations, both in UCT and in an informative PUCT variant.

A PUCT case with budget eight and all six root actions visited is also documented, in which the chosen action is worth (0.70) against an alternative of (0.95).

These records are cited as archived observations, not as experiments independently reproduced in this article.

Visiting all root options does not mean that their continuations have been evaluated exhaustively, and increasing the budget does not on its own guarantee a monotone growth in the quality of each selection.

The direct application of Theorem 1 to a comparison obtained in any of these searches additionally requires that the alternatives be represented by the appropriate MIN/MAX model and domain of completions.

The theorem can ground a retrospective certification if adequate evidence exists; it does not automatically attribute that knowledge to the algorithm at the moment it chose.

In Responsible AI, the contribution of the framework is not to automatically promise fairness, transparency, or legal compliance.

It is to offer a precise discipline for formulating evidential claims: what has occurred in an execution, what relation is deduced from the evidence, and what has been attested through a procedure whose soundness is justified.

This discipline can serve as a basis for an audit, but its practical effects must be evaluated separately.

8.3 The distinction among selection, demonstration, and certification grounds an epistemic requirement of RAI

Responsible AI (RAI) comprises development, evaluation, and oversight practices aimed at making explicit the reasons, limits, and responsibility of decisions assisted by computational systems. The results of the present article can contribute to a discipline of evidential attribution:

- to distinguish the record of a selection, the evidence that determines a comparison, and the certificate whose acceptance has been justified.

For an audit, this suggests documenting the objective used, the relevant legal alternatives, the observed evidence, the assumptions about unknown values, the provenance of the certificates, and the limits of their verification.

It also allows us to identify overclaims: a success label does not on its own establish soundness; a preference under one heuristic does not establish superiority under another objective function; a pairwise comparison is not equivalent to global optimality.

These are possible and conditional applications, not validation results in health-care, administrative, or industrial settings.

Theorem 1 does not on its own establish fairness, absence of bias, effective transparency, regulatory compliance, or legal responsibility; each property would require specific normative standards, evidence, and verifications.

The historical delimitation deserves the same rigor as the assumptions of the theorem.

Bound-based search techniques, the monotonicity of MIN/MAX, and the properties of lattice functions have specific antecedents.

The contribution should be presented through the exact formulation of the problem, the semantics of compatible completions, and the demonstrative construction set out here, without presupposing historical priority before it has been exhaustively checked.

8.4 Antecedents, delimitation of originality, and organization

The use of bounds in search has antecedents such as B* and solver-based search methods (Berliner 1979; Winands, Björnsson, and Saito 2008; Cazenave and Saffidine 2010).

MIN/MAX expressions belong to the theory of lattice polynomial functions, which includes known algebraic properties of monotonicity and thresholds (Goodstein 1967; Couceiro and Marichal 2012).

The proof set out here is self-contained, but no historical priority of the mathematical characterization is claimed without a specific comparison with the literature.

After the semantic definitions, the auxiliary logical results, the constructive lemma and Theorem 1, its verification instance, and the conditions of formal certification are presented.

The final discussion considers, in a strictly conditional manner, possible uses in the audit of UCT and PUCT, formal verification, and Responsible AI, distinguishing the proven results from their ethical and moral consequences that cannot be derived exclusively from the formulas.

8.5 Background: the formal limits of computation

The historical perspective does not aim to equate this research with the undecidability theorems or to turn it into a new general limit of computability.

Its methodological affinity consists in demanding a rigorous separation between executing a procedure and proving a proposition about what that procedure allows us to establish.

Research on the limits of what a machine can establish has one of its fundamental antecedents

in the work of Alan Turing, On Computable Numbers, with an Application to the Entscheidungsproblem (1936).

Turing showed that the rigorous definition of a mechanical procedure makes it possible not only to identify what can be computed, but also to prove the nonexistence of general procedures capable of solving certain classes of problems.

The epistemic importance of this contribution lies in the fact that the limits of computation can become the object of mathematical proof.

It is not enough to observe that a procedure produces results or fails in certain cases: it is necessary to establish what can be guaranteed about its behavior and under what conditions.

Our research situates itself in that methodological tradition, although it addresses a different problem.

We do not claim to prove new general limits of computability or decidability. We examine what can legitimately be inferred from an algorithmic selection and when the available evidence allows us to demonstrate the strict superiority of one alternative over another.

- The first article formally established that an observed selection does not, by itself, entail a demonstrated strict preference.
- The present work advances from that limit of inference toward a positive characterization: under explicit mathematical assumptions, it determines the necessary and sufficient conditions for establishing strict superiority even when part of the information remains unknown.

The question that links both works is therefore epistemic:

What authorizes us to transform the result of a computational operation into a demonstrated claim?

The answer does not depend solely on a machine having produced a decision, but on the logical and mathematical relations that can be established between the available evidence and the conclusion that is to be justified.

The question that has guided the article is not which alternative a machine chose, but what evidence authorizes us to assert that this alternative is superior.

The path followed allows us to answer by distinguishing the limits of inference from a selection, the mathematical conditions of a universal comparison, and the additional obligations of any computational certification.

9 Conclusions: proven results and epistemic significance

The first conclusion is logical: selection, Resolved Comparison, and Demonstrated Preference are not equivalent notions.

A recorded selection does not by itself entail demonstrated preference in the specified class of structures; a resolved comparison may end in a tie or favor the rival.

Moreover, the existential definitions do not exclude admissible witnesses from disagreeing: their agreement requires a condition independent of the standard.

These distinctions delimit the legitimate inference from a trace, without turning the absence of proof into a proof of falsity.

The second conclusion is the central mathematical result.

Under H1, for finite MIN/MAX expressions, one alternative strictly exceeds another in all compatible completions if and only if its exact lower extremum is greater than the exact upper extremum of the rival.

Two directions are distinguished: sufficiency follows from chaining valid bounds; necessity is

shown by a common completion constructed by threshold clipping.

The latter does not require the expressions to use disjoint variables.

Assumption H2 of the earlier texts is thereby eliminated as a requirement of the strengthened equivalence; dependencies among distinct variables remain outside the product domain when they impose additional constraints.

The third conclusion concerns informational sufficiency.

In the twelve-leaf tree, five known values suffice to establish

$$L_{E^+}(a_5) = 0.95 > -0.30 = U_{E^+}(a_0)$$

and, with it, a relation that is true for all compatible completions.

After removing one observation, two completions respect the remaining four values and produce opposite results.

It is thus shown, for the constructed instance, that determining a comparison does not require determining all the values.

The five observations are inclusion-minimal for that certificate, not necessarily globally minimal.

The fourth conclusion delimits the scope of optimality judgments.

Superiority over one alternative does not establish superiority over all; unique optimality requires coverage of the legal rivals under the same objective.

Under H1 and MIN/MAX, the characterization by exact extrema also admits a necessary and sufficient formulation for that uniqueness.

At another level, a single strictly superior legal witness refutes the universal correctness of a configuration over a domain that includes it, but does not determine the error frequency of an algorithm.

The fifth conclusion concerns formal verification.

The theorem establishes a semantic truth within a mathematical class, not the automatic correctness of a piece of software.

An accepted certificate supports that truth only when the verifier's soundness has been established, together with the conditions of provenance, typing, identity of objective, and admissibility of witnesses.

The determination of a comparison may precede its computational recognition; nor has it been shown that a positive delay or a cost advantage always exists.

Finally, these results are relevant to the retrospective audit of UCT and PUCT decisions and to Responsible AI practices, provided that the relevant models and standards are declared. The historical search examples are configured observations, not consequences of Theorem 1 nor a general demonstration of the incorrectness of such methods.

The epistemic conclusion is thereby delimited:

a choice identifies an action produced; a demonstration identifies a relation that follows from the evidence; a valid certification further requires guarantees about the procedure that recognizes it.

The exact historical priority of the algebraic characterization and the correctness of specific implementations continue to require independent verification. The preceding results will allow us to raise the ethical dimension without confusing the different types of justification.

A conclusion valid under an objective function answers a mathematical question; the legitimacy of the objective, the inclusion of the relevant alternatives, and the consequences of the decision also respond to normative questions.

The existence of a proof does not eliminate those questions: it makes them more visible by delimiting precisely what the calculation does establish.

10 Ethical and moral implications of algorithmic attribution

As we have indicated, the formal result has implications for the way in which decisions of computational systems are communicated and justified, but these implications must not be confused with new normative theorems.

Whoever asserts that a system has chosen "the best" alternative must clarify what is meant by "best": according to what objective, against which alternatives, and with what evidence.

The mathematical, logical, and philosophical work shows why a recorded selection, a favorable comparison, and an established universal conclusion are claims of different logical strength.

The ethical requirement to communicate that difference honestly, by contrast, derives from additional normative criteria of responsibility and not from the MIN/MAX operation alone.

It is therefore necessary to distinguish superiority relative to an objective function from the moral justification of the decision.

- The former requires an evaluation and a proof in accordance with the declared assumptions.
- The latter requires reasons concerning the value of the objective, the interests affected, and the relevant obligations.

A correctly optimized action may still require justification external to the optimization model.

10.1 Correctness relative to an objective and moral justification

An action may correctly maximize a declared objective function and yet not be morally justified under an independent normative framework.

For a class of models that admits that separation, the following scheme of normative non-entailment can be expressed:

$$Correct_V(d) \not\models_{\mathcal{K}_{V,N}} MorallyJustified_N(d).$$

where (d) is an action or decision; Correct$_{\mathbf{V}}$(d) asserts its correctness relative to the objective function (V); MorallyJustifiedN(d) expresses its justification under the normative framework (N); and $\mathcal{K}$V,N is a class of structures that includes at least one model in which correctness according to (V) does not satisfy the conditions of justification of (N).

This statement is presented as a scheme of normative non-entailment conditioned on the declared class of models, not as a new consequence of Theorem 1.

A conceptual countermodel illustrates that separation:

- if (V) values exclusively cost reduction, it might regard as optimal a decision that restricts an essential service;
- a normative framework (N) that requires protecting access to that service might not regard it as morally justified.

Optimization according to (V) and justification under (N) thus respond to different criteria.

The example is illustrative and does not establish empirical facts or a specific legal characterization.

Nor does it show that every optimized action is morally wrong; not justified is not automatically equivalent to morally prohibited.

The distinction prevents transferring, without argument, a superiority within an objective function to a judgment about duties, justice, or rights.

Even when a comparison is determined with all the mathematical precision required in its model, the question may remain open whether the adopted objective was adequate, whether the relevant alternatives were correctly delimited, and whether the decision respected obligations external to the calculation.

Epistemic responsibility begins with not attributing to the system more knowledge than the evidence establishes. Whoever communicates the results must separate what is observed from what is deduced, what is certified from what is merely accepted by a program, and what is undetermined from what is demonstrated false.

10.2 Responsibility for the evidence, uncertainty, and human oversight

The most direct ethical contribution is a discipline of responsibility in the attribution of conclusions. Whoever designs, deploys, or audits a system (the human) should differentiate between:

- Observed results.
- Formally demonstrated guarantees.
- Merely presumed properties.

Accepting software without a justification of soundness, presenting an approximate bound as an exact extremum, or presenting a heuristic comparison as truth under another objective are forms of overclaiming whose logical structure can be identified with the proposed framework.

For this reason, communicating uncertainty also requires distinguishing three statuses:

- Determined relation.
- Refuted relation.
- Undetermined relation.

Indeterminacy does not on its own oblige us to suspend all action, but it does prevent presenting as demonstrated a superiority that remains open in the compatible models.

From this perspective, human oversight makes epistemic sense when it allows us to review:

- The objective.
- The evidence.
- The assumptions.
- The provenance of the certificates.
- The scope of the certificates.

Oversight does not acquire that sense when it is limited to accepting a label produced by the very system being evaluated.

These considerations can guide Responsible AI applications, but they do not constitute empirical proof of effects on:

- Human behavior.
- Fairness.

- Safety.
- Legitimacy of the decision.
- Compliance with specific norms.

Those consequences require ethical premises, empirical evaluations, and legal frameworks that Theorem 1 does not provide.

The philosophical contribution consists in specifying where a mathematical proof ends and where the responsibility to justify the criteria and uses of a decision begins.

- A machine can select.
- A proof can justify a relation between alternatives.
- A certification can attest that proof only if the corresponding procedure satisfies the required guarantees.

None of these operations, by itself, resolves the moral question of which ends should be pursued or who answers for their choice. Precisely because we know how to delimit what the calculation proves, we can formulate with greater rigor what human responsibility must justify.

10.3 Human responsibility in the face of the limits of algorithmic demonstration

The philosophical contribution of this research consists in establishing a **fundamental distinction between what a computational operation produces, what the evidence allows us to demonstrate, and what requires an independent ethical justification.**

The mathematical characterization developed in this work allows us to delimit, under explicit assumptions, the conditions under which a comparison can be regarded as demonstrated. However, the demonstration of superiority with respect to an objective function does not by itself establish the moral legitimacy of that objective.

This distinction has implications that transcend the strictly computational sphere:

- **In the sphere of decisions that affect other people**, it requires distinguishing between the result of an algorithmic procedure and the reasons that may justify its use in those decisions. The intervention of a computational system must not be confused with an automatic transfer of responsibility.
- **In the legal sphere**, it provides an epistemic perspective relevant to examining the evidence, the traceability, and the justifications of automated decisions. The determination of concrete obligations and responsibilities corresponds, however, to the applicable legal frameworks.
- **In the scientific and technological sphere**, it allows us to differentiate the formal correctness of a procedure from the legitimacy of the objectives for which it is designed, and to recognize that a mathematical guarantee does not on its own demonstrate respect for fundamental rights.
- **In the sphere of human responsibility**, it reminds us that oversight cannot be reduced to accepting the results of a machine: it must be able to examine the ends pursued, the premises used, the available evidence, and the consequences of the decisions.

The contribution of this research does not consist in mathematically proving a moral theory, but in specifying an epistemic boundary that is indispensable for properly formulating ethical questions. A system can execute a procedure correctly and reach an optimal result according to the criteria assigned to it, without that resolving whether those criteria are just, legitimate, or morally acceptable.

From this perspective, the following statements can be distinguished:

Computational efficiency does not replace ethical justification; mathematical correctness is not equivalent to moral legitimacy; and the existence of an algorithmic decision does not eliminate human responsibility for the ends and consequences of its use.

Precisely because we can delimit rigorously what the calculation demonstrates, we can identify what remains to be justified. This delimitation preserves the space of human judgment about ends, consequences, and freedom of choice.

That distinction is essential so that technological development does not confuse the capacity to decide with the moral authority to do so.

While Turing examined the difficulties of interpreting the behavior of machines, this work formulates a logical-formal distinction between a computational output and the conclusions justified by the evidence.

In his 1950 article, "Computing Machinery and Intelligence," Turing pointed out that machines frequently surprise those who do not calculate enough, and that the opposite view is due to a fallacy:

"as soon as a fact is presented to a mind all consequences of that fact spring into the mind simultaneously with it." (Turing 1950, 451).

Turing's reflection shows that the consequences of a fact are not automatically evident to the one who observes it. At a different logical level, our countermodel shows that selection, by itself, does not entail a demonstrated strict preference.

Within a declared model, observing a selection does not suffice to attribute a demonstrated preference, and there exists an exact criterion that separates what the evidence determines from what it leaves open.

Whoever attributes a demonstrated preference without an admissible witness, under a declared standard, asserts more than has been demonstrated.

What still requires an independent normative grounding is why the agents who design, deploy, or use AI systems are obliged to respect these distinctions and under what conditions.

This question does not invalidate the mathematical proofs presented: it delimits the step from an epistemic theory to a complete normative theory of RAI.

Several lines of research remain open: extending the characterization to dependent variables and non-monotone functions, formally examining the correctness of verifiers, and testing the results in real audits.

Finally, the research establishes general results of justification and refutation whose scope does not depend on a particular algorithm recognizing them during its execution. Before enumerating them, it is worth recalling the mathematical assumption that delimits the central characterization.

The central mathematical characterization is proved under Assumption 1 (H1), called independence of the unknown values. H1 is a mathematical assumption of the model: each unknown variable can take any value in its closed, bounded, nonempty real interval, independently of the other unknown variables. It is not statistical independence. The observed values remain fixed and, if one and the same variable appears several times, all its occurrences receive the same value in each completion. H1 allows all admissible combinations of the unknown values to be considered and provides a sufficient condition for proving the equivalence of Theorem 1, without claiming that it is necessary in all possible models.

Within their respective assumptions and classes of structures, the article formulates five claims of general scope:

- **An algorithmic selection does not by itself entail a demonstrated strict preference.** The observation of a choice does not authorize that inference without the additional

conditions of evidence required by the declared standard.

- **A relation is semantically determined when it is true in all completions compatible with the evidence.** This is the defining condition of determination, formulated through semantic entailment.
- **Separation of exact extrema is a necessary and sufficient condition of strict dominance.** Theorem 1 proves this equivalence for any pair of finite MIN/MAX expressions evaluated under H1, even when the alternatives share variables. MIN/MAX denotes the minimum and maximum operations, used, among other areas, in the minimax evaluation of decision trees.
- **Consistent extension of the evidence preserves previously determined relations.** This property holds when the semantics remains fixed and the set of completions compatible with the new evidence is nonempty.
- **A single valid counterexample suffices to refute the universal correctness of an algorithm.** The refutation refers to the declared domain, configuration, and objective function. It is not necessary to find multiple errors to refute a universal claim, although a single case likewise does not allow its frequency to be established.

Semantic determination and the monotonicity of the evidence do not need H1 by themselves; this assumption intervenes specifically in the proof of the necessary and sufficient characterization of Theorem 1.

The archived records of UCT ("Upper Confidence Bounds Applied to Trees," a tree-search method based on confidence bounds; Kocsis and Szepesvári 2006) and PUCT (a variant that incorporates prior probabilities to guide exploration; Rosin 2011; see its applications in Silver et al. 2017, 2018) illustrate the application of the principle of refutation by counterexamples. Their evidential force with respect to the configurations examined depends on the validation of the records and of the reference evaluations. They do not show that these methods are incorrect in general, nor that AlphaZero (an artificial intelligence system that combines neural networks with tree search; Silver et al. 2018) is. It is necessary to distinguish the formal proof of a general property from the experimental verification of a specific implementation.

The scope of these results is logical and mathematical, always relative to the declared conditions and domains. It does not imply that all algorithms are incorrect or that all their decisions can be evaluated using MIN/MAX; it does, however, allow us to specify which conclusions the evidence justifies and which require additional proofs.

The contribution of this work consists in distinguishing what a system produces, what the evidence allows us to demonstrate, and what a certification procedure can validly attest. This distinction offers a logical-mathematical foundation for the epistemic and evidential dimension of Responsible AI (RAI), without equating computational correctness with the ethical or moral justification of a decision.

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